

Which Privacy Blanket is Optimal in the Shuffle Model?

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Abstract

In recent years, the shuffle model has emerged as a prevalent paradigm in privacy-preserving data analysis, centered on the principle of *privacy amplification via shuffling*: an individual user’s report is obscured by the “background noise” of other users’ messages, a phenomenon intuitively known as the *privacy blanket* [BBGN19]. This paper initiates a foundational and systematic study of this mechanism from an information-theoretic perspective. We investigate the following optimal noise-design problem: given that a target user’s message Y_1 follows a distribution P , what is the optimal blanket noise distribution Q that maximizes privacy? Specifically, when $Y_1 \sim P$ and the remaining messages $Y_2, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} Q$ are shuffled to produce an output $\mathbf{Z} = (Y_{\sigma(i)})_{i=1}^n$, we seek the Q that affords the strongest protection for Y_1 under various metrics, including mutual information $I(Y_1; \mathbf{Z})$, total-variation-information $I_{\text{TV}}(Y_1; \mathbf{Z})$, message recovery advantage, and expected posterior variance.

Our analysis reveals a series of non-intuitive results that challenge the conventional heuristic of setting $Q = P$. First, we prove that the optimal noise distribution Q generally *deviates* from the target distribution P . For binary alphabets, we show that the (near-)uniform distribution is optimal in a strong sense. For general finite alphabets, we derive an explicit analytical form $Q \propto \sqrt{P(1-P)}$ that achieves asymptotic optimality for mutual information. Furthermore, we demonstrate that our analytical framework transcends the shuffle model, yielding new security insights into broader cryptographic primitives such as the *ideal cipher model* and *honey encryption*.

Finally, we extend our results to the shuffle-DP paradigm, where messages $Y_i = \mathcal{R}(X_i)$ are outputs of ε_0 -locally differentially private mechanisms. We establish a new, tight information-theoretic upper bound $I(X_1; \mathbf{Z} \mid (X_i)_{i=2}^n) \leq \frac{(e^{\varepsilon_0/2} - 1)^2}{2n} + \mathcal{O}(n^{-3/2})$. This result provides a sharp characterization that matches the optimal privacy-amplification parameters known in the literature [FMT22], while offering a novel interpretation of the shuffling gain.

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1 Introduction

1.1 Problem Setting

We investigate a fundamental architecture in privacy-preserving data analysis: the *single-message shuffle model*. The problem is formally defined as follows: n users, each holding an input X_i drawn from a distribution P_i , submit a message $Y_i = g(X_i)$ to a common collection point. Let $\mathbf{Z} = (Y_{\sigma(i)})_{i=1}^n$ denote the shuffled sequence produced by a uniformly random permutation σ , and let $K = \sigma^{-1}(1)$ represent the position of the first user’s message in the output¹. Our central inquiry is to characterize the information leakage of X_1 and K revealed by the observed shuffled sequence $\mathbf{Z}$.

This framework unifies two important scenarios: when g is the identity function, it characterizes the classical *anonymous voting* mechanism prevalent in practice; when g is a localized differentially private (ϵ_0 -LDP) mechanism, it corresponds to the *single-message shuffle DP model* [BEM⁺17, BBGN19, CSU⁺19], a paradigm that has recently gained prominence for its capacity to provide “privacy amplification by shuffling” [EFM⁺19, BBGN19, FMT22, SCW26b]. Furthermore, this setup serves as a core building block for privacy analysis of various protocols in the *multi-message shuffle model* [CZ21, LWY22, LLF⁺24], which we discuss further in Section 2.4.

Even in the *shuffle-only* model (where $Y_i = X_i$) and assuming Y_i are mutually independent, the analysis remains highly non-trivial. A tempting but flawed intuition suggests that the probability of identifying the target message Y_1 from the shuffled sequence $\mathbf{Z}$ is bounded by the “oblivious” baseline of $1/n$. Although the index K is marginally uniform over $[n]$ by construction, its posterior distribution $K|\mathbf{Z}$ becomes skewed whenever the underlying distributions $\{P_i\}$ are non-identical. Consequently, an adversary may correctly identify the target $Y_1 = Z_K$ with a probability strictly greater than $1/n$.

More fundamentally, the leakage of Y_1 encompasses more than a mere point estimate of K ; it fundamentally captures the adversary’s belief update based on the statistical ensemble $\mathbf{Z}$. This requires a rigorous analysis of the posterior distribution of Y_1 conditioned on $\mathbf{Z}$.

While K remains uniformly distributed when all Y_i are i.i.d., the posterior becomes remarkably complex when messages follow heterogeneous distributions P_i . Nevertheless, a sharp analytical solution with a clear combinatorial structure emerges in what we define as the **canonical setting**: a scenario where a single target message $Y_1 \sim P$ is “submerged” in a “blanket” of $n - 1$ i.i.d. samples drawn from a noise distribution Q [BBGN19, SCW26a].

More remarkably, the general case involving heterogeneous distributions P_i can be reduced, without loss of generality, to the aforementioned canonical setting. Suppose there exists a constant $\gamma \in [0, 1]$ such that each P_i admits the decomposition:

$$P_i = \gamma Q^{\text{com}} + (1 - \gamma) \text{LO}_i,$$

where Q^{com} represents the common component shared across all distributions, and LO_i denotes the respective “left-over” residues. Let $\mathbf{Z}^r = \mathcal{S}(\{Y_1\} \cup \{Y'_i\}_{i=2}^{N+1})$ be a reduced sequence where $N \sim \text{Bin}(n - 1, \gamma)$, $Y'_i \stackrel{\text{i.i.d.}}{\sim} Q^{\text{com}}$, and $\mathcal{S}$ denotes a uniformly random permutation. The data-processing inequality ensures that the leakage of Y_1 and X_1 in the original sequence $\mathbf{Z}$ is strictly bounded by that in its “reduced” counterpart $\mathbf{Z}^r$ [BBGN19, FMT22].

This framework unifies several known techniques: when Q^{com} is taken as the maximal common component of all P_i , we recover the so-called *blanket decomposition* [BBGN19]; when $Q^{\text{com}} = P_1$, it yields a variant of the *clone decomposition* [FMT22, SCW26a]. While the number of messages in

¹In this notation, $\sigma(i) = j$ implies that the message at the i -th output position originates from the j -th user.

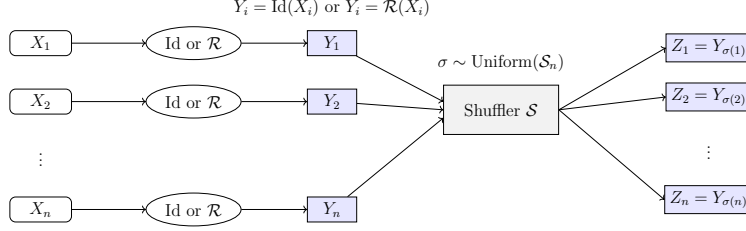


Figure 1: Illustration of the single-message shuffle model. Each user i generates a local message Y_i , either deterministically ($Y_i = \text{Id}(X_i) = X_i$) or via a local randomizer ($Y_i = \mathcal{R}(X_i)$). The shuffler then applies a random permutation σ , uniformly sampled from the permutation group $\mathcal{S}_n$, to produce the anonymized outputs $Z_i = Y_{\sigma(i)}$.

the raw blanket reduction, $1 + N$, is stochastic [BBGN19], it can be regularized to a fixed size n by introducing a dummy symbol $\perp$ outside the support of P_i [SCW26b]. Let $\mathbb{1}_\perp$ be the single-point distribution on $\perp$. By defining an augmented distribution $\bar{Q} = \gamma Q^{\text{com}} + (1 - \gamma)\mathbb{1}_\perp$, the reduced sequence $\mathbf{Z}^r$ becomes equivalent to a standard n -message sequence $\bar{\mathbf{Z}} = \mathcal{S}(\{Y_1\} \cup \{Y'_i\}_{i=2}^n)$ with $Y'_i \stackrel{\text{i.i.d.}}{\sim} \bar{Q}$. This technique effectively “regularizes” the stochastic sample size N into a fixed-length vector of size n , thereby streamlining the mathematical treatment of divergence and information metrics [SCW26b].

We remark that this canonical setting in the shuffle-only model is not only analytically tractable and universally reducible, but also directly characterizes several fundamental problems in computer security.

1.2 Broader Background and Application

1.2.1 Honeyword

Beyond its role in privacy protection, the shuffle model also captures the essence of “honey” techniques in computer security. A representative example is the *honeyword* mechanism, a password-leakage detection scheme designed to mitigate the impact of frequent password breaches [JR13].

In a honeyword-based system, the authentication server stores a user’s true password along with $n - 1$ plausible decoys—called *honeywords*—and then shuffles them, thereby forcing the adversary to contend with a set of n plausible candidates [JR13]. If an adversary compromises the password database, they obtain a set of n indistinguishable candidates rather than the true password alone. When a login attempt is made using one of these passwords, the system can detect misuse if a honeyword is entered and raise an alert [JR13, WCW⁺18].

Contemporary honeyword generation [WCW⁺18, WZD⁺22, CLL⁺22] typically draws decoys i.i.d. from a trained password probability model [WAMG09, DAC⁺15, MUS⁺16]. Let P represent the distribution of human-chosen passwords, and Q the distribution used to generate honeywords. This framework establishes a rigorous equivalence between honeyword security and the canonical shuffle-only model: the genuine password $Y_1 \sim P$ is “camouflaged” by $n - 1$ honeywords acting as blanket noise Q .

1.2.2 Ideal Cipher (IC) and Honey Encryption (HE)

To abstract away the complexities of specific symmetric-key primitives, the Ideal Cipher (IC) model assumes that for each key $k \in \mathcal{K}$, the encryption algorithm $\text{Enc}_k^{\text{IC}, \mathcal{M}}$ is a uniform random

permutation over the message space $\mathcal{M}$. This abstraction is widely adopted in the formal security analysis of cryptographic protocols, e.g., [BPR00, BRS02, CPS08, BS23].

Consider a target message M sampled from a distribution P . A computationally unbounded adversary, upon intercepting a ciphertext, may perform a brute-force search over the key space $\mathcal{K}$. This exhaustive process yields a set of candidate plaintexts consisting of the true message $Y_1 \sim P$ and $|\mathcal{K}| - 1$ spurious decryptions. In a standard IC, these spurious decryptions are effectively uniform samples from $\mathcal{M}$. When P significantly deviates from the uniform distribution, the adversary can leverage this statistical discrepancy to distinguish the true message and recover the secret key. Such exhaustive attacks are particularly devastating when the key entropy is low, such as in password-based encryption [CBJR15, GBD16, CLW⁺21].

To mitigate this, *Honey Encryption* (HE) [JR14, JRT16] introduces a *distribution-transforming encoder* to ensure that “honey” messages (decryptions under incorrect keys) follow a specific distribution Q . Specifically, let $\text{encode} : \mathcal{M} \rightarrow \mathcal{M}'$ be a one-to-many encoder satisfying $\text{encode}(m) \cap \text{encode}(m') = \emptyset$ for all $m \neq m'$, with a corresponding decoder $\text{decode} : \mathcal{M}' \rightarrow \mathcal{M}$. The HE scheme is defined as:

$$\text{Enc}_k^{\text{HE}}(m) = \text{Enc}_k^{\text{IC}, \mathcal{M}'}(\text{encode}(m)), \quad \text{Dec}_k^{\text{HE}}(c) = \text{decode}(\text{Dec}_k^{\text{IC}, \mathcal{M}'}(c)).$$

Under the IC model, the spurious decryptions in HE follow the distribution $\text{decode}(U_{\mathcal{M}'})$, where $U_{\mathcal{M}'}$ is the uniform distribution on $\mathcal{M}'$. By employing the inverse transform sampling technique, $\text{decode}(U_{\mathcal{M}'})$ can be configured to approximate any desired distribution Q on $\mathcal{M}$ with a quantization error vanishing as the message space $|\mathcal{M}'|$ increases. The core design philosophy of HE is to choose $Q = P$ (within precision) such that the resulting set of candidates—one $Y_1 \sim P$ and $|\mathcal{K}| - 1$ samples from Q —provides no distinguishing advantage to the adversary.

Notably, when the secret keys of IC/HE are uniformly distributed over $\mathcal{K}$, the security analysis of these models becomes formally equivalent to the canonical shuffle-only model (with $n = |\mathcal{K}|$) investigated in this paper.

1.2.3 The Optimal Blanket Problem

These observations naturally lead to a fundamental research question: given a target distribution P , what is the optimal “blanket” noise distribution Q that minimizes the information leakage of the true message? In particular, does the optimal Q necessarily coincide with P , as a naive symmetry argument might suggest? Our main result provides a definitive negative answer to this question, demonstrating that the “intuitive” choice $Q = P$ is generally sub-optimal.

1.3 Characterizing Optimal Blanket in the Shuffle-Only Model

When designing the optimal blanket noise Q for a specific input distribution P , the standard framework of Differential Privacy (DP, see Section 2.1) becomes insufficient, as its worst-case guarantees are designed to be distribution-agnostic. To capture the fine-grained, distribution-dependent leakage inherent in the shuffle model, we evaluate privacy through the lens of four classical metrics:

1. **Information-Theoretic Metrics:** (Shannon) Mutual Information (MI) and Total-Variation-Information (TV-Information), both of which belong to the family of f -information derived from f -divergence [PW22].
2. **Operational/QIF Metrics:** Message Recovery Advantage (MRA) and the Expected Posterior Variance (EPV), which fall under the framework of Quantitative Information Flow (QIF, see Section 2.2).

We derive either asymptotic or non-asymptotic analytical expressions for these metrics. Our analysis reveals several counter-intuitive properties regarding optimal message (Y_1) privacy²:

- **Mutual Information:** We provide a clean asymptotic characterization demonstrating that for a general (non-uniform) P , the optimal noise distribution follows $Q \propto \sqrt{P(1-P)}$, rather than the intuitive choice $Q = P$. Notably, for a binary alphabet, the optimal Q is exactly a uniform distribution regardless of the bias in P .
- **TV-Information:** We derive analytical expressions for general P . For the binary case, we obtain a closed-form solution showing that the optimal distribution is “quasi-uniform”—specifically, $(\frac{\lfloor n/2 \rfloor}{n}, \frac{\lceil n/2 \rceil}{n})$ or its permutation.
- **Message Recovery Advantage & Expected Posterior Variance:** MRA characterizes the success rate gain of a Bayesian adversary, while EPV provides a natural measure of posterior uncertainty via L_2 error for ordered domains. We provide analytical characterizations for both metrics and employ numerical optimization to determine the Q that minimizes these leakage bounds.

Conversely, for position (K) privacy, we provide a distinct asymptotic characterization:

$$\lim_{n \rightarrow \infty} I(K; \mathbf{Z}) = D_{\text{KL}}(P \| Q), \quad \lim_{n \rightarrow \infty} I_{\text{TV}}(K; \mathbf{Z}) = D_{\text{TV}}(P, Q).$$

This highlights a fundamental **trade-off** between message and position privacy: the noise distribution Q optimized to minimize content-based leakage typically proves sub-optimal for protecting the source’s position, and vice versa.

1.3.1 Mutual Information

From a classical information-theoretic perspective, privacy leakage is quantified by the *mutual information* between X and Y . Let $H(X) = -\sum_x p_X(x) \log p_X(x)$ denote the Shannon entropy. The mutual information between these two random variables is defined as:

$$I(X; Y) = H(X) - H(X | Y) = \sum_{x,y} p_{X,Y}(x,y) \log \left(\frac{p_{X,Y}(x,y)}{p_X(x)p_Y(y)} \right).$$

Throughout this paper, all logarithmic terms, including the Shannon entropy $H(P) = -\sum_i p_i \ln p_i$, are taken as natural logarithms. Consequently, all information-theoretic quantities are measured in nats, ensuring more concise analytical expressions.

Theorem 1. *Consider the canonical shuffle-only model setting where the target user’s message $Y_1 \sim P$ and $n-1$ blanket messages i.i.d. according to Q . If $P \ll Q$, i.e., $\text{Supp}(P) \subseteq \text{Supp}(Q)$, then the mutual information between the shuffle position and the shuffled output satisfies*

$$I(K; \mathbf{Z}) = D_{\text{KL}}(P \| Q) - \frac{\chi^2(P \| Q)}{2n} + O(n^{-3/2}), \quad n \rightarrow \infty,$$

where D_{KL} is the Kullback-Leibler divergence, $\chi^2(P \| Q) = \sum_y \frac{(P(y) - Q(y))^2}{Q(y)}$.

²While $Y_1 = X_1$ holds in the shuffle-only model, we present these results in terms of message privacy, as they extend naturally to the shuffle-DP setting (see Section 7).

Theorem 2. *In the canonical setting of the shuffle-only model, when $P \ll Q$, the mutual information between the targeted message and the shuffled output satisfies*

$$I(Y_1; \mathbf{Z}) = \frac{\sum_i \frac{p_i - p_i^2}{q_i}}{2n} + O(n^{-3/2}), \quad n \rightarrow \infty,$$

where p_i and q_i denote the probability mass functions of P and Q , respectively.

Theorem 3. *In the canonical setting of the shuffle-only model, for a fixed P , the choice of Q that asymptotically minimizes $I(Y_1; \mathbf{Z})$ satisfies $q_i \propto \sqrt{p_i(1 - p_i)}$.*

Beyond its intrinsic theoretical value, these results offer crucial insights into the security of Honey Encryption. In the large- n limit (where n corresponds to the key space size), the leakage regarding the secret key K converges to a constant $D_{\text{KL}}(P \| Q)$, representing an irreducible “information floor”. Meanwhile, for any Q such that $P \ll Q$, the mutual information $I(Y_1; \mathbf{Z})$ vanishes asymptotically.

More remarkably, our analysis quantifies an inherent dichotomy between positional anonymity (identifying the key) and message confidentiality (recovering the content). Let $m = |\text{supp}(P)|$ denote the support size of P . While setting $Q = P$ achieves perfect position secrecy ($I(K; \mathbf{Z}) = 0$), it incurs a message leakage coefficient of $m - 1$. In contrast, the optimum Q^* minimizes message leakage at the cost of providing a constant-rate side channel regarding the key index K .

Example 1.1. *We conduct numerical experiments using a Zipf distribution $P = \text{Zipf}(m, \alpha)$ with $m = 4$ and $\alpha = 0.7$ ($p_i \propto i^{-\alpha}$), and a uniform distribution Q over $[4] = \{1, 2, 3, 4\}$. Given that closed-form expressions for $I(K; \mathbf{Z})$ and $I(Y_1; \mathbf{Z})$ are analytically intractable, we estimate these values via Monte Carlo simulations.³ As illustrated in Fig. 2a, the empirical estimates of mutual information exhibit tight alignment with the theoretical asymptotic curves.*

Next, we empirically validate Theorem 3 using the same $P = \text{Zipf}(4, 0.7)$ setup. We compare the baseline $Q_1 = P$ against the theoretically optimal choice Q_2 , where $q_i \propto \sqrt{p_i(1 - p_i)}$. The resulting Monte Carlo curves and their corresponding theoretical predictions are displayed in Fig. 2b. For this configuration, the constants are $C_1 = m - 1 = 3$ and $C_2 = \left(\sum_i \sqrt{p_i(1 - p_i)}\right)^2 \approx 2.81$. The empirical results are in high agreement with our theoretical findings; notably, the transition from $C_1 = 3$ to $C_2 \approx 2.81$ yields a 6.3% reduction in asymptotic leakage.

1.3.2 Total-Variation-Information

Total Variation (TV) distance is a classical metric between two probability distribution defined by $D_{\text{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|$. TV-information [PW22] between two random variables Y_1 and $\mathbf{Z}$ is defined as the expected TV distance between the prior P and the posterior distribution of Y_1 given $\mathbf{Z}$:

$$I_{\text{TV}}(Y_1; \mathbf{Z}) = \mathbb{E}_{\mathbf{Z}}[D_{\text{TV}}(P, \text{Law}(Y_1 | \mathbf{Z}))].$$

TV-information quantifies the expected deviation of the adversary’s posterior belief from the prior P . It is intrinsically linked to the notion of semantic security in cryptographic settings [TW23].

Theorem 4. *In the canonical setting in the shuffle-only model, if $P \ll Q$, then the total-variation-information satisfies:*

$$\lim_{n \rightarrow \infty} I_{\text{TV}}(K; \mathbf{Z}) = D_{\text{TV}}(P, Q).$$

³Specifically, we compute $I(K; \mathbf{Z}) = \mathbb{E}_{\mathbf{Z}} \left[\sum_{k=1}^n \Pr(K = k | \mathbf{Z}) \log(n \Pr(K = k | \mathbf{Z})) \right]$. To ensure statistical convergence, expectations are approximated by the sample mean across 10^5 independent realizations of $\mathbf{Z}$.

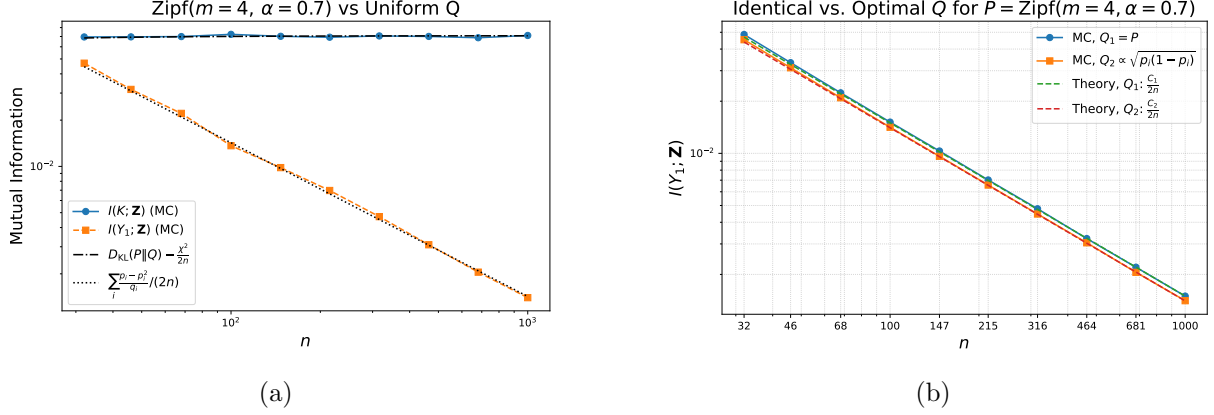


Figure 2: Exact vs. asymptotic MI in the canonical setting and empirical validation of the optimal distribution Q .

Theorem 5. *In the canonical shuffle-only model, when Q is supported on $[m] = \{1, 2, \dots, m\}$ and $P \ll Q$, the TV-information admits the following exact combinatorial representation:*

$$I_{TV}(Y_1; \mathbf{Z}) = \frac{1}{2} \sum_{i=1}^m p_i \mathbb{E}_{\vec{n} \sim \text{Multi}(n, Q)} \left[\left| \frac{n_i}{nq_i} - \sum_{j=1}^m p_j \frac{n_j}{nq_j} \right| \right].$$

Theorem 6. *In the canonical setting in the shuffle model, when the support size is two, for any $n \in \mathbb{N}$, the optimal Q (i.e., $\arg \min_Q I_{TV}(Y_1; \mathbf{Z})$) has a form $(\frac{\lfloor n/2 \rfloor}{n}, \frac{\lceil n/2 \rceil}{n})$ or $(\frac{\lceil n/2 \rceil}{n}, \frac{\lfloor n/2 \rfloor}{n})$.*

These theorems establish an asymptotic parallel between TV-information and Shannon mutual information, both exhibiting a constant bound for positional leakage. Crucially, however, for binary distributions, the optimality of the quasi-uniform distribution under TV-information holds strictly even in the non-asymptotic regime.

Example 1.2. *Consider a ternary alphabet with a prior distribution $P = (0.2, 0.3, 0.5)$. For a fixed sample size $n = 10$, we evaluate the total-variation-information $I_{TV}(Y_1; \mathbf{Z})$ by varying the parameters (q_2, q_3) of the blanket noise distribution $Q = (1 - q_2 - q_3, q_2, q_3)$. Figure 3 illustrates the landscape of I_{TV} as a function of the noise configuration. Notably, the global minimum leakage is achieved at $(q_2, q_3) \approx (0.30, 0.40)$, yielding an I_{TV} of 0.167. This numerically optimized Q provides a strictly stronger privacy guarantee than the standard heuristic of setting $Q = P$ ($q_2 = 0.3, q_3 = 0.5$), which results in a higher information leakage of 0.175.*

1.3.3 Message Recovery Advantage and Expected Posterior Variance

The message recovery advantage quantifies the marginal gain of a Bayesian adversary in identifying the true value of Y_1 after observing the shuffled output. It is defined as the increase in the maximum posterior probability relative to the prior:

$$\text{Adv}_{MR}(Y_1; \mathbf{Z}) := \mathbb{E}_{\mathbf{z} \sim \mathbf{Z}} [\max_y \Pr[Y_1 = y | \mathbf{Z} = \mathbf{z}]] - \max_y \Pr[Y_1 = y].$$

This metric serves as a pivotal security benchmark for Honeywords [WCW⁺18, JR13] and Honey Encryption [JR14], and has been recently adopted in the context of Shuffle Differential Privacy [JGAP23].

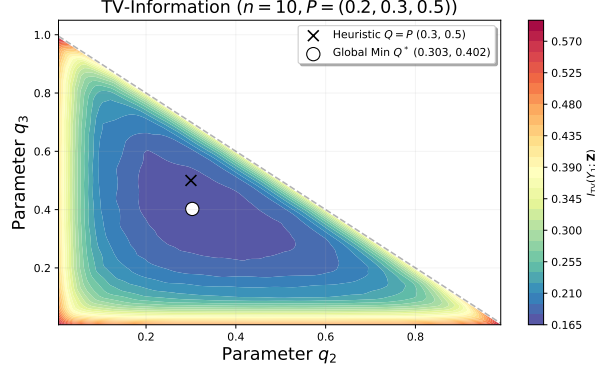


Figure 3: Optimal distribution $Q = (1 - q_2 - q_3, q_2, q_3)$ for minimizing TV-information leakage.

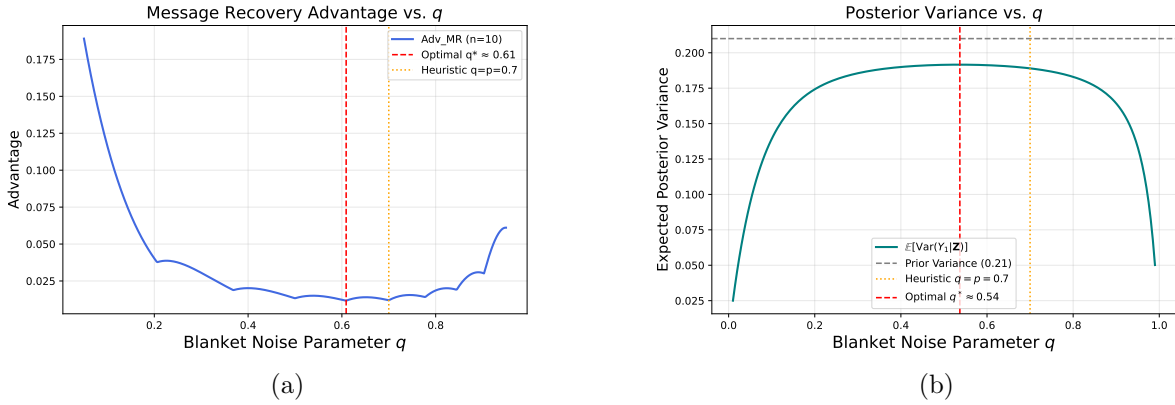


Figure 4: Optimal distribution $Q = (1 - q, q)$ minimizing MRA and EPV ($P = (0.3, 0.7)$ on $\{0, 1\}$ and $n = 10$).

For numerical datasets, privacy leakage is naturally characterized by the reduction in an adversary’s estimation error. Whereas the expected empirical variance of n i.i.d. samples from P follows the classical $(1 - 1/n)\text{Var}_P$ relation, the expected posterior variance within the canonical shuffle model exhibits a significantly more intricate dependence when the noise distribution Q deviates from P .

We derive analytical expressions for both metrics of the canonical shuffle-only model in Section 5. Our numerical investigations demonstrate that the optimal Q generally deviates from P .

Example 1.3. *To validate the impact of the blanket noise, we consider a binary source $P = (0.3, 0.7)$ with $n = 10$ and evaluate both metrics as functions of the noise parameter q (where $Q = (1 - q, q)$). As shown in Fig. 4, the results confirm that the “probability-matching” heuristic ($q = 0.7$) is sub-optimal for both metrics.*

Specifically, the message recovery advantage achieves its minimum at $q \approx 0.61$, yielding a value of 0.0117 (compared to 0.0123 at $q = 0.7$). Correspondingly, the expected posterior uncertainty—measured by the variance—reaches its maximum of 0.192 at $q \approx 0.54$, outperforming the 0.189 achieved by the standard heuristic.

1.4 Information-Theoretic Privacy Amplification for Shuffle DP

While the shuffle-only analysis reveals that a shared common component among user distributions can provide provable privacy, the underlying assumptions—independence of X_i and the existence of a substantial blanket—are often too restrictive for practical deployment. To ensure worst-case security, the single-message shuffle DP model typically requires each user to process their input X_i through an ε_0 -LDP local randomizer $\mathcal{R}$ prior to shuffling [BBGN19, CSU⁺19]. This architecture effectively homogenizes the blanket noise by leveraging the inherent privacy of $\mathcal{R}(X_i)$, thereby providing robust protection even against arbitrary input correlations.

In the canonical shuffle DP setting, the shuffler outputs $\mathbf{Z} = \mathcal{S}(\{Y_1\} \cup \{Y'_i\}_{i=2}^n)$, where $Y_1 = \mathcal{R}(X_1)$ with $X_1 \sim P$, and the Y'_i are i.i.d. draws from a noise distribution Q .

For convenience, $\Pr[\mathcal{R}(x) = y]$ is denoted by $\mathcal{R}_x(y)$. Let $\bar{P} = \sum_{x \in \mathbb{X}} P(x) \mathcal{R}_x$ denote the marginal distribution of Y_1 . While the position leakage $I(K; \mathbf{Z})$ is minimized when $Q = \bar{P}$, we show that the asymptotically optimal Q for minimizing the input leakage $I(X_1; \mathbf{Z})$ is given by:

$$Q(y) \propto \sqrt{\sum_{x \in \mathbb{X}} P(x) (\mathcal{R}_x(y) - \bar{P}(y))^2}.$$

This result admits a crisp interpretation: the optimal noise $Q(y)$ is proportional to the standard deviation of the local randomizer's output probability at y with respect to the input distribution $x \sim P$. Notably, this recovers our optimal $Q(y) \propto \sqrt{P(y)(1 - P(y))}$ for the shuffle-only model by setting $\mathcal{R} = \text{Id}$ (where $\mathcal{R}_x(y) = \mathbb{1}[x = y]$). A more detailed discussion is provided in Section 6.2.3

Although this optimal Q depends on both the randomizer $\mathcal{R}$ and the target distribution P , requiring external injection, we prove that the natural blanket distribution $\bar{Q}^B$ (definition in Section 6.2.1)—contributed inherently by other users—is remarkably effective. By applying this reduction, we derive information-theoretic privacy bounds against an strong adversary who possesses auxiliary knowledge of $\mathbf{X}_{-1}$ (or the entire $\mathbf{X}$ for positional leakage). Crucially, these bounds are distribution-agnostic and hold without any assumptions regarding the independence of the users' inputs.

Let $\mathcal{P}(\mathbb{X}^n)$ denote the set of all joint distributions on the database $\mathbf{X} = (X_1, \dots, X_n)$, allowing arbitrary dependence. For any $P_{\mathbf{X}} \in \mathcal{P}(\mathbb{X}^n)$, let $I_{P_{\mathbf{X}}}(\cdot)$ denote mutual information computed under the joint law induced by $P_{\mathbf{X}}$, the local randomizer $\mathcal{R}$, and the shuffler. Our main bounds establish that if the local randomizer $\mathcal{R}$ satisfies ε_0 -LDP, then the following inequalities hold uniformly over all joint distributions $P_{\mathbf{X}} \in \mathcal{P}(\mathbb{X}^n)$ ⁴:

$$I_{P_{\mathbf{X}}}(K; \mathbf{Z} \mid \mathbf{X}) \leq 2\varepsilon_0, \quad I_{P_{\mathbf{X}}}(X_1; \mathbf{Z} \mid \mathbf{X}_{-1}) \leq \frac{(e^{\varepsilon_0/2} - 1)^2}{2n} + O(n^{-3/2}).$$

These results offer a novel characterization of privacy amplification. Notably, the coefficient $\frac{(e^{\varepsilon_0/2} - 1)^2}{2}$ is proven to be optimal under the blanket decomposition framework [SCW26b].

Discussion. [FMT22] has established that the overall mechanism in the shuffle DP model satisfies (ε, δ) -DP with $\varepsilon = O\left(\frac{(1 - e^{-\varepsilon_0})}{\sqrt{n}} \sqrt{e^{\varepsilon_0} \log(1/\delta)}\right)$. They prove that the dependence on this parameter is tight. Specifically, for a fixed δ , this implies $\varepsilon = O(\varepsilon_0/\sqrt{n})$ when $\varepsilon_0 < 1$, and $\varepsilon = O(e^{\varepsilon_0/2}/\sqrt{n})$ when $\varepsilon_0 > 1$. By Pinsker's inequality, our mutual information bound yields $I_{\text{TV}}(X_1; \mathbf{Z} \mid \mathbf{X}_{-1}) \leq \frac{e^{\varepsilon_0/2} - 1}{2\sqrt{n}} + o(\frac{1}{\sqrt{n}})$, where the term $e^{\varepsilon_0/2} - 1$ scales as $O(\varepsilon_0)$ for $\varepsilon_0 < 1$ and as $O(e^{\varepsilon_0/2})$ for $\varepsilon_0 > 1$.

⁴In particular, the first inequality implies the unconditional bound $I_{P_{\mathbf{X}}}(K; \mathbf{Z}) \leq 2\varepsilon_0$ because K is independent of $\mathbf{X}$.

Table 1: Summary of mutual information bounds in the shuffle model.

Setting	Mutual Information $I(K; \mathbf{Z})$	Mutual Information $I(X_1; \mathbf{Z})$
Canonical Shuffle-Only ($P = Q$)	0 (Property 1)	$\frac{m-1}{2n} + O(n^{-3/2})$ (Theorem 10)
Canonical Shuffle-Only ($P \ll Q$)	$D_{\text{KL}}(P\ Q) - \frac{\chi^2(P\ Q)}{2n} + O(n^{-3/2})$ (Theorem 1)	$\frac{\sum_i \frac{p_i - q_i^2}{q_i}}{2n} + O(n^{-3/2})$ (Theorem 2)
Canonical Shuffle-Only ($P \not\ll Q$)	Theorem 19	Theorem 20
General Shuffle-Only (heterogeneous)	Theorem 14	Theorem 9
Shuffle-DP (ε_0 -LDP then shuffle)	$I(K; \mathbf{Z} \mid \mathbf{X}) \leq 2\varepsilon_0$ (Theorem 13)	$I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1}) \leq \frac{(\varepsilon_0/2 - 1)^2}{2n} + O(n^{-3/2})$ (Theorem 17)

Note: n is the population size; m denotes the support size of P ; $D_{\text{KL}}(P\|Q) = \sum_i p_i \log(\frac{p_i}{q_i})$ and $\chi^2(P\|Q) = \sum_i \frac{(p_i - q_i)^2}{q_i}$ are the Kullback–Leibler and chi-squared divergences of P relative to Q , respectively; $P \ll Q$ means $\text{Supp}(P) \subseteq \text{Supp}(Q)$. The constants hidden in $O(\cdot)$ may depend on P , Q , or $\mathcal{R}$.

It is important to clarify that the (ε, δ) -DP amplification theorem alone cannot derive our mutual information bound, as the potential failure probability δ precludes any universal privacy guarantee. Conversely, our mutual information bound only implies a bound on TV-information—indicating that the posterior distribution of X_1 is likely close to its prior—but does not encompass the (ε, δ) amplification result. Nevertheless, the privacy amplification of mutual information provides a clear, unified explanation for these piecewise behaviors, identifying $e^{\varepsilon_0/2} - 1$ as the fundamental underlying parameter.

It should be clarified that our bounds evaluate the mutual information conditioned on $\mathbf{X}_{-1}$. Such conditioning is indispensable in the mutual information analysis of DP mechanisms [CY16]; since DP does not safeguard the global information of the database, it cannot prevent the leakage of X_1 when it is strongly correlated with the rest of the inputs $\mathbf{X}_{-1}$. Remark 3 illustrates this phenomenon.

1.5 Technical Overview

Analysis of Canonical Shuffle-Only Model. The analytical tractability of the canonical shuffle-only model stems from a fundamental relationship between the probability distribution of $\mathbf{Z} = \mathcal{S}(\{Y_1 \sim P\} \cup \{Y'_i \sim Q\}_{i=2}^n)$ and that of $\mathbf{Z}' = \mathcal{S}(\{Y'_i \sim Q\}_{i=1}^n)$:

$$\forall \mathbf{z}, \frac{\Pr[\mathbf{Z} = \mathbf{z}]}{\Pr[\mathbf{Z}' = \mathbf{z}]} = \frac{1}{n} \sum_{i=1}^n \frac{P(z_i)}{Q(z_i)}, \quad \Pr[K = i \mid \mathbf{Z} = \mathbf{z}] = \frac{\frac{P(z_i)}{Q(z_i)}}{\sum_{j=1}^n \frac{P(z_j)}{Q(z_j)}}.$$

This relationship admits a clear intuitive interpretation and a formal combinatorial proof: the posterior distribution of the index K is proportional to the likelihood ratio at each position, while the ratio between the distributions of $\mathbf{Z}$ and $\mathbf{Z}'$ is equivalent to the average likelihood ratio across all positions⁵.

This property was previously leveraged by [BBGN19] to derive privacy amplification bounds in shuffle DP, and by [SCW26a] to calculate the expected success probability of re-identification attacks in the shuffle model. In this work, we further extend this technique to derive non-asymptotic

⁵As $n \rightarrow \infty$, the distributions of $\mathbf{Z}$ and $\mathbf{Z}'$ converge, indicating that the information of Y_1 is effectively masked by the blanket noise.

analytical expressions for TV-information, message recovery advantage, and expected posterior variance.

Regarding the bounds for mutual information, since the KL divergence involves logarithmic functions, its non-asymptotic expression cannot be easily simplified. Consequently, we employ Taylor expansions and high-order moment control inequalities (such as the Rosenthal inequality) to obtain its asymptotic characterization.

Analysis of Canonical Shuffle DP Model. We establish a powerful lemma that characterizes the asymptotic principal term of $I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1})$ in the canonical shuffle DP setting. The resulting coefficient carries a clear information-theoretic interpretation: it represents a form of “variance” of the χ^2 -divergence between the local randomizer $\mathcal{R}_x$ and the blanket noise Q , where x is drawn from P .

Lemma 1 (Mutual information for one signal mixed with s i.i.d. Q -samples). *Let $\mathbb{Y}$ be a finite output alphabet, and assume $Q(y) > 0$ for all $y \in \mathbb{Y}$. For an integer $s \geq 1$, let $\mathbf{Z} = \mathcal{S}(\{Y_1\} \cup \{Y'_i\}_{i=1}^s)$ where $Y_1 = \mathcal{R}(X_1)$, $Y'_i \stackrel{i.i.d.}{\sim} Q$. Define the corresponding chi-square divergences:*

$$\chi^2(\mathcal{R}_x \| Q) := \sum_y \frac{\mathcal{R}_x(y)^2}{Q(y)} - 1, \quad \overline{\chi^2} := \mathbb{E}_{x \sim P}[\chi^2(\mathcal{R}_x \| Q)].$$

Let $\bar{P}(y) = \sum_x P(x)\mathcal{R}_x(y)$ be the P -mixture of the $\mathcal{R}_x$'s (i.e., the marginal distribution of Y_1). Then the mutual information between X_1 and $\mathbf{Z}$ satisfies, for all sufficiently large s ,

$$I(X_1; \mathbf{Z}) = \frac{\overline{\chi^2} - \chi^2(\bar{P} \| Q)}{2(s+1)} + O((s+1)^{-3/2}). \quad (1)$$

The optimal distribution Q that minimizes the residual term $\overline{\chi^2} - \chi^2(\bar{P} \| Q)$ is derived via the Cauchy-Schwarz inequality (Theorem 16). Furthermore, we establish a tight upper bound for the resulting leakage coefficient in the practical case where Q is chosen as the generalized blanket distribution $\bar{Q}^B$ of the local randomizer $\mathcal{R}$.

Lemma 2. *Let $\mathcal{R} : \mathbb{X} \rightarrow \mathbb{Y}$ be an ε_0 -DP local randomizer and $\bar{Q}^B$ be a generalized blanket distribution on $\mathbb{Y}$. For a distribution P on $\mathbb{X}$, let $\bar{P}$ denote the marginal distribution on $\mathbb{Y}$ defined by $\bar{P}(y) = \sum_{x \in \mathbb{X}} P(x)\mathcal{R}_x(y)$. Then, the following inequality holds:*

$$\mathbb{E}_{x \sim P}[\chi^2(\mathcal{R}_x \| \bar{Q}^B)] - \chi^2(\bar{P} \| \bar{Q}^B) \leq (e^{\varepsilon_0/2} - 1)^2.$$

Furthermore, this bound is tight.

In fact, Lemma 1 also yields *lower bounds* on the mutual information in the shuffle-DP setting. When $X_2, X_3, \dots, X_n$ are i.i.d., the output distribution $Q = \mathcal{R}(X_2)$ naturally plays the role of an *effective blanket distribution*. This interpretation is illustrated by Example 6.1.

1.6 Related Work

While Section 2 provides a broader overview of DP, Shuffle DP, and MI-based privacy metrics and QIF, this section focuses on the literature directly underpinning our technical contributions. In the canonical shuffle-only setting where $Q = P$, the shuffled output $\mathbf{Z}$ consists of n i.i.d. samples from P . Under this scenario, since the posterior of K remains uniform, the posterior distribution

of Y_1 reduces to the empirical distribution of $\mathbf{Z}$. Consequently, the four privacy metrics analyzed in this work recover several fundamental results from classical probability and information theory.

Specifically, let $\hat{P}_n$ denote the empirical distribution of n i.i.d. samples from P with support size m . A well-known result establishes that the expected KL divergence behaves as $\mathbb{E}[D_{\text{KL}}(\hat{P}_n \| P)] = \frac{m-1}{n} + o(\frac{1}{n})$ [PW22]. For the Total Variation (TV) distance, it is established that $\mathbb{E}[D_{\text{TV}}(\hat{P}_n, P)] = \Theta(\sqrt{\frac{m-1}{n}})$ [HJW15]. Regarding the message recovery advantage, when P is uniform, the maximum posterior probability problem maps to the classical balls-into-bins maximum load problem. For n balls and m bins, the maximum load scales as $\frac{n}{m} + \Theta(\sqrt{\frac{n \log m}{m}})$ as $n \rightarrow \infty$, which has been leveraged in prior work to show that MRA is $\Theta(\sqrt{\frac{\log m}{mn}})$ [JGAP23]. Finally, for the Expected Posterior Variance, standard statistical theory shows that the expected variance of the empirical distribution is $(1 - \frac{1}{n})\text{Var}(P)$.

Building upon these foundations, this paper extends the analysis to the generalized canonical setting where $Q \neq P$, providing new theoretical insights and characterizations of these privacy leakage metrics.

1.7 Contributions

Our main contributions are summarized as follows:

- **Optimal Noise Design in Canonical Shuffling:** We provide a comprehensive security analysis of the shuffle model across four fundamental privacy metrics. We prove that the “probability-matching” heuristic ($Q = P$) is generally sub-optimal and derive closed-form expressions for the optimal noise configuration Q^* , notably for mutual information and binary sources.
- **MI-based Privacy Amplification:** We establish a novel mutual information privacy amplification bound for the shuffle-DP framework. Our analysis highlights the quantity $e^{\varepsilon_0/2} - 1$ as the pivotal driver of privacy gain, offering a new information-theoretic perspective on the classical privacy amplification theorems previously established in the (ε, δ) -DP setting.
- **Bridging Shuffle Model and Cryptography:** We bridge the single-message shuffle model with classical cryptographic models, including the IC model and Honey Encryption. By quantifying the one-shot information leakage from ciphertexts to keys and messages, we identify a structural trade-off that characterizes the relationship between these two leakage types.

1.8 Paper Organization

The remainder of this paper is organized as follows. Section 2 provides the necessary background on differential privacy and the shuffle model, while situating our work within the broader context of mutual-information-based privacy and quantitative information flow. Sections 3, 4, and 5 develop our theoretical analysis of the canonical shuffle-only setting using these distinct privacy metrics: mutual information, TV-information, and QIF (i.e., MRA and EPV), respectively. Section 6 extends this analysis to the shuffle-DP model. Further implications and potential extensions are addressed in Section 7, and Section 8 concludes the paper with a discussion of several open problems.

For clarity, the main body of the paper focuses on proof sketches and core conceptual insights; the complete technical proofs and detailed arguments are deferred to the Appendices.

2 Preliminaries

2.1 Differential Privacy

Differential privacy provides a principled way to quantify the privacy of randomized algorithms. Informally, an algorithm satisfies differential privacy if its output distribution is nearly unchanged when the data of a single individual is modified. In this way, the published outcome reveals very limited information about any particular participant.

The standard (central) model of differential privacy assumes the existence of a trusted curator that holds the raw data of all n users and applies a DP mechanism on top of it [Dwo06, DR14]. In many practical deployments, however, the *local* model is preferred: under *local differential privacy* (LDP), each user perturbs their data on the client side before sending it to the collector, thereby removing the need to trust the server [DKY17, DPT17, CJK⁺18, YGZ⁺24]. The price of this weaker trust assumption, though, is typically a noticeable degradation in utility because the noise must be added per user [DJW13, BS15].

A line of work therefore considers the *shuffle model*, which inserts a shuffling component between the clients and the analyzer [BEM⁺17, CSU⁺19, EFM⁺19]. The shuffler anonymizes the users' reports by randomly permuting them before they reach the server, and this additional anonymity can be leveraged to improve accuracy. In this paper we concentrate on the *single-message shuffle model*, where each user sends exactly one randomized report and these reports are shuffled prior to aggregation [FMT23]. Here we give the formal definitions of DP, LDP, and shuffle DP.

Definition 1 (Hockey-Stick Divergence). *The hockey-stick divergence between two random variables P and Q is defined as:*

$$D_\alpha(P \parallel Q) = \int \max\{0, P(x) - \alpha Q(x)\} dx,$$

where we use the notation P and Q to refer to both the random variables and their probability density functions.

We say that random variables P and Q are (ε, δ) -indistinguishable if for all set T :

$$\Pr[P \in T] \leq e^\varepsilon \Pr[Q \in T] + \delta.$$

If two datasets X and X' have the same size and differ only by the data of a single individual, they are referred to as neighboring datasets (denoted by $X \simeq X'$).

Definition 2 (Differential Privacy). *An algorithm $\mathcal{R} : \mathbb{X}^n \rightarrow \mathbb{Z}$ satisfies (ε, δ) -differential privacy if for all neighboring datasets $X, X' \in \mathbb{X}^n$, $\mathcal{R}(X)$ and $\mathcal{R}(X')$ are (ε, δ) -indistinguishable.*

Definition 3 (Local Differential Privacy). *An algorithm $\mathcal{R} : \mathbb{X} \rightarrow \mathbb{Y}$ satisfies local (ε, δ) -differential privacy if for all $x, x' \in \mathbb{X}$, $\mathcal{R}(x)$ and $\mathcal{R}(x')$ are (ε, δ) -indistinguishable.*

Here, ε is referred to as the privacy budget, which controls the privacy loss, while δ allows for a small probability of failure. When $\delta = 0$, the mechanism is also called ε -DP.

Following conventions in the shuffle model based on randomize-then-shuffle [BBGN19, CSU⁺19], we consider a single-message protocol defined by a local randomizer $\mathcal{R} : \mathbb{X} \rightarrow \mathbb{Y}$, where $\mathbb{Y}$ is the *message space*. Given a dataset $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{X}^n$, each user i applies the local randomizer to produce a message $y_i = \mathcal{R}(x_i)$. The messages are then passed through a shuffler $\mathcal{S} : \mathbb{Y}^n \rightarrow \mathbb{Y}^n$, which outputs a uniformly random permutation of its inputs. Denoting by $\mathcal{R}^n(\mathbf{x}) = (\mathcal{R}(x_1), \dots, \mathcal{R}(x_n))$, the overall randomized mechanism in the shuffle model is $\mathcal{M}(\mathbf{x}) = \mathcal{S} \circ \mathcal{R}^n(\mathbf{x})$. Any subsequent analysis can be viewed as post-processing of this shuffled output.

Definition 4 (Differential Privacy in the Shuffle Model). *A mechanism $\mathcal{M} = \mathcal{S} \circ \mathcal{R}^n$ satisfies (ε, δ) -differential privacy in the shuffle model if for all neighboring datasets $X, X' \in \mathbb{X}^n$, the distributions $\mathcal{S} \circ \mathcal{R}^n(X)$ and $\mathcal{S} \circ \mathcal{R}^n(X')$ are (ε, δ) -indistinguishable.*

Privacy Amplification via Shuffling. The privacy amplification theorem in the single-message shuffle model postulates that when n users independently apply an ε_0 -LDP mechanism to randomize their data, the shuffled output distribution satisfies $(\varepsilon(\varepsilon_0, \delta, n), \delta)$ -DP. For sufficiently large n and non-negligible δ , it holds that $\varepsilon(\varepsilon_0, \delta, n) \ll \varepsilon_0$ [FMT23]. Specifically, a sequence of increasingly refined bounds has been established: $\varepsilon(\varepsilon_0, \delta, n) = O(\varepsilon_0 \sqrt{\frac{\log(1/\delta)}{n}})$ (restricted to $\varepsilon_0 < 1$) [EFM⁺19], $\varepsilon(\varepsilon_0, \delta, n) = O(\min\{1, \varepsilon_0\} \cdot e^{\varepsilon_0} \sqrt{\frac{\log(1/\delta)}{n}})$ [BBGN19], and $\varepsilon(\varepsilon_0, \delta, n) = O((1 - e^{-\varepsilon_0}) \sqrt{\frac{e^{\varepsilon_0} \log(1/\delta)}{n}})$ [FMT22]. This privacy amplification effect is universal: a simple shuffling step significantly enhances the privacy guarantees of any underlying LDP mechanism.

2.2 Quantative Information Flow

Quantitative Information Flow (QIF) aims to characterize the refinement of an adversary’s *inferential capability* regarding a secret input upon observing the output of a mechanism. A distinguishing feature of the QIF framework is its explicit dependence on the prior distribution of the input; it defines “privacy leakage” as the gain in the adversary’s *optimal inference success rate* (or its generalized utility) [ACPS12, AAC⁺15]. For a comprehensive treatment and discussion of the QIF framework, we refer the reader to the monograph [ACM⁺20].

Formal Framework: Channels, Priors, and Gains Let the secret input be a random variable $X \in \mathbb{X}$ with a prior distribution $\pi \in \mathcal{P}(\mathbb{X})$. A (potentially randomized) mechanism can be abstracted as a *channel* $C : \mathbb{X} \rightarrow \mathbb{Y}$, defined by the conditional probabilities $C[y | x] = \Pr[Y = y | X = x]$, where $Y \in \mathbb{Y}$ is the output observable by the adversary.

To unify various adversarial objectives, we introduce a *gain function* $g : \mathbb{X} \times \mathcal{A} \rightarrow \mathbb{R}$, where $\mathcal{A}$ is the set of possible actions available to the adversary. The value $g(x, a)$ represents the gain attained by the adversary when the true secret is x and the chosen action is a .

Prior/Posterior Vulnerability and g -leakage Given a prior π and a gain function g , the adversary’s optimal expected gain before observing any output is termed the *prior vulnerability*:

$$V_g(\pi) = \max_{a \in \mathcal{A}} \sum_{x \in \mathbb{X}} \pi(x) g(x, a).$$

Upon observing an output $Y = y$, the adversary’s belief is updated to the posterior distribution $\pi(\cdot | y)$, yielding an optimal expected gain of $\max_{a \in \mathcal{A}} \sum_x \pi(x | y) g(x, a)$. Taking the expectation over all possible outputs $y \in \mathbb{Y}$ yields the *posterior vulnerability*:

$$V_g(\pi, C) = \sum_{y \in \mathbb{Y}} \Pr[Y = y] \cdot \max_{a \in \mathcal{A}} \sum_{x \in \mathbb{X}} \pi(x | y) g(x, a).$$

Definition 5. $\mathcal{L}_g^+(\pi, C) = V_g(\pi, C) - V_g(\pi)$ is the additive g -leakage with respect to C under prior distribution π .

This paper focuses on two canonical instantiations of the gain function, catering to different data types:

- **One-try Secret Guessing:** For discrete random variables, we consider the maximum success probability in a single inference attempt. This is formally defined by $\mathcal{A} = \mathbb{X}$ and $g(x, a) = \mathbb{1}[x = a]$. Under this gain function, the posterior vulnerability $V_g(\pi, C)$ corresponds to the *Bayes vulnerability*, representing the adversary’s optimal probability of correctly identifying the secret in one shot.
- **Negative L_2 Loss:** For continuous random variables supported on $\mathbb{X} \subseteq \mathbb{R}$, we characterize the adversary’s precision in estimating a numerical value. This is modeled by the negative squared error loss, where $\mathcal{A} = \mathbb{R}$ and $g(x, a) = -(x - a)^2$. In this setting, the adversary’s optimal action a is the posterior mean $\mathbb{E}[X \mid Y = y]$, and the posterior vulnerability reflects the expected posterior variance of the secret.

The QIF framework, under the two specific gain functions discussed above, provides the formal definitions and operational interpretations for the privacy metrics adopted in this paper: *message recovery advantage* and *expected posterior variance*.

Discussion. [JGAP23] initiated the exploration of the message recovery advantage within the shuffle model. Notably, they compared the MRA across LDP, shuffle-only, and combined LDP-plus-shuffle settings. Their findings indicate that, against an uninformed adversary (who possesses no prior knowledge of individual data and assumes a uniform prior over datasets), the shuffling process provides the predominant privacy gain for the k -Randomized Response mechanism (see definition in Example 6.1). However, a general analytical treatment of MRA remains technically formidable, with closed-form expressions previously restricted to a few specialized cases.

Specifically, while [JGAP23] focuses on an uninformed adversary, conventional privacy research typically assumes a strong (informed) adversary who knows all individuals’ data except that of the target user. Although they provide an analytical characterization of MRA for k -RR against an uninformed adversary, the MRA for other mechanisms remains largely unexplored. Furthermore, in the presence of a strong adversary, even the simple k -RR mechanism becomes analytically intractable; consequently, prior work only offered brief numerical discussions and left a rigorous analysis for future investigation.

Our work addresses these gaps in two key aspects. First, in the shuffle-only model, we derive exact analytical expressions for the MRA in the canonical setting, which generalize their prior results (where $P = Q$ is restricted to the uniform distribution). Second, within the shuffle DP model, we establish a hierarchical implication: our mutual information bound yields a total-variation-information bound, which in turn controls the message recovery advantage—owing to the fact that TV distance upper-bounds the deviation between maximum posterior probabilities. This theoretical chain establishes that for any ϵ_0 -LDP mechanism under shuffling, the MRA scales as $O\left(\frac{e^{\epsilon_0/2}-1}{\sqrt{n}}\right)$, providing a universal guarantee against strong adversaries.

2.3 Mutual-Information-Based Privacy

A fundamental problem in privacy-preserving data analysis is how to *quantify* information leakage (see surveys [WE18, HMB⁺21, BGY⁺21, UO23]). Beyond differential privacy [Dwo06], alternative frameworks have been proposed from the perspectives of *quantitative information flow* (QIF) [ACPS12, AAC⁺15, ACM⁺20] and *mutual information* [dPCF12, MF13, CY16].

The information-theoretic privacy measure from the literature is defined as follows [CY16]:

Definition 6 (Mutual Information-Differential Privacy). *A randomized mechanism $P_{Y|X^n}$ satisfies ϵ -mutual-information differential privacy if*

$$\sup_{i, P_{X^n}} I(X_i; Y | X^{-i}) \leq \epsilon \text{ nats.}$$

Classic (ϵ, δ) -DP and MI-DP are mutually transferable [CY16]:

- ϵ -DP implies $\min\{\epsilon, \epsilon^2\}$ -MI-DP: This follows from the fact that bounded likelihood ratios provide an upper bound on the Kullback-Leibler divergence.
- ϵ -MI-DP implies $(0, \sqrt{2\epsilon})$ -DP: This relationship is established via Pinsker’s inequality.

[CY16] advocates for the MI-DP perspective due to several key advantages:

1. Natural Composition Properties: The additive nature of mutual information facilitates a more intuitive understanding of the composition properties inherent in classical DP.
2. Explication of the Strong Adversary Assumption: The MI-DP definition explicitly formalizes the “strong adversary” assumption—where the attacker knows all individual data except for the target’s—which is often only implicitly defined in standard DP. This clarity helps prevent potential misunderstandings and misapplications. For instance, as highlighted in [KM11], when correlations exist among X_i , standard DP mechanisms may fail to provide sufficient privacy protection.

Discussion. A substantial line of work has investigated the relationships between DP and MI-based privacy. Several studies have established formal connections between *central* DP and MI [CY16]. Others have examined the relationship between *local* DP and MI [MMP⁺10, WHW⁺16, DJW13, NöiA24]. However, to the best of our knowledge, the link between the *shuffle* model of DP and MI-based privacy has not been explicitly characterized prior to this work.

Our work derives the asymptotic expressions for both message mutual information and positional mutual information within the canonical setting. Furthermore, we characterize the optimal configuration of the blanket noise distribution Q tailored to a target message distribution P , thereby enriching the scope and depth of MI-based privacy research. Within the shuffle DP framework, we establish a privacy amplification bound under the MI-DP definition. Beyond its intrinsic theoretical value, this bound provides an intuitive explanation for the amplification effects observed in classical (ϵ, δ) -DP. Aligning with the perspectives in [CY16], we believe and anticipate that the MI-based privacy lens will continue to offer profound insights for future research in privacy protection.

A related effort [GGK⁺21] derived specialized mutual information bounds in the shuffle model to establish lower bounds on estimation error and sample complexity; however, its objectives and analytical techniques differ fundamentally from ours.

2.4 Multi-Message Shuffle Model

In the multi-message shuffle model, each user contributes multiple messages to the shuffler, which uniformly permutes all received messages before releasing the resulting collection. Current protocol designs within this model can be broadly categorized into two distinct paradigms.

The first category focuses on enhancing privacy through the injection of auxiliary *blanket noise* messages (e.g., [CZ21, LWY22, LLF⁺24]). Our proposed framework is natively compatible with this setting. A canonical example is the private histogram estimation protocol in [CZ21]: in addition

to submitting the output of an ε_0 -DP RAPPOR mechanism [EPK14] applied to their input, each user contributes k independent samples drawn from an input-independent distribution.

In contrast to the single-message shuffle model, where blanket noise—defined as the common component of the local randomizers’ output distributions—is contributed probabilistically by each user, the multi-message shuffle model permits the direct injection of a specific quantity and distribution of additional blanket noise. This shifts the selection of the optimal distribution Q from a theoretical abstraction to a design problem of heightened practical significance and immediate utility.

The second paradigm utilizes the multi-message shuffle model as a fundamental primitive for Secure Multi-Party Computation (MPC). The objectives and methodologies here differ fundamentally from those of privacy amplification or blanket noise. In this line of research, security is typically formalized within the simulation-based paradigm of MPC rather than through information-theoretic metrics such as mutual information. Consequently, our framework does not extend to these protocols. For instance, the “split-and-mix” protocol [IKOS06] allows users to send additive secret shares of their inputs to the shuffler, achieving statistically secure summation. Recent advancements have further demonstrated that, assuming all users are honest, the multi-message shuffle model supports one-round, non-interactive, and information-theoretically secure MPC for arbitrary functions [HIKR23, Hiw25, BEG25, BEGI25]. Notably, compared to the first paradigm, this approach often necessitates stricter trust assumptions and incurs substantially higher communication overhead.

2.5 Posterior Distributions under the Canonical Setting

The following theorem characterizes how an observer of the shuffled sequence $\mathbf{Z}$ updates its belief about both the position (K) of the distinguished user’s message and the message itself (Y_1).

Theorem 7. *Consider the canonical shuffle-only model setting where the target user’s input $Y_1 \sim P$ and $n - 1$ blanket messages i.i.d. according to Q . Conditioning on the shuffled output $\mathbf{Z} = \mathbf{z} = (z_1, z_2, \dots, z_n)$, the posterior distributions of K and Y_1 are given by*

$$\Pr[K = k \mid \mathbf{Z} = \mathbf{z}] = \frac{\frac{P(z_k)}{Q(z_k)}}{\sum_{i=1}^n \frac{P(z_i)}{Q(z_i)}},$$

$$\Pr[Y_1 = y \mid \mathbf{Z} = \mathbf{z}] = \sum_{k=1}^n \Pr[K = k \mid \mathbf{Z} = \mathbf{z}] \mathbb{1}[z_k = y].$$

The derivation of the posterior distribution of K follows the approach in prior work [WCW⁺18, SCW26a]; for completeness, the detailed proof is provided in Appendix A. The expression for the posterior of the message value $Y_1 = Z_K$ follows directly from the law of total probability.

Notably, if for some i , $Q(z_i) = 0$ but $P(z_i) > 0$, we interpret $P(z_i)/Q(z_i) = \infty$, indicating that z_i must have come from P with certainty.

Our analysis primarily focuses on the regime where P is absolutely continuous with respect to Q . Intuitively, if there exists a symbol y^* such that $P(y^*) > 0$ but $Q(y^*) = 0$, the occurrence of $Y_1 = y^*$ in the shuffled sequence $\mathbf{Z}$ would be uniquely attributable to the first user, leading to non-vanishing leakage even as $n \rightarrow \infty$. Conversely, when $P \ll Q$, the information leakage typically vanishes in the large- n limit. To substantiate this intuition, we provide a formal analysis of mutual information leakage for the case where $P \not\ll Q$ in Theorems 19 and 20.

Interpretation. Theorem 7 demonstrates that, upon observing the shuffled sequence $\mathbf{Z}$, the posterior probability that the first user’s message occupies position k is proportional to the likelihood ratio $\frac{P(z_k)}{Q(z_k)}$. In particular, when $P = Q$, this ratio equals one for all k , resulting in a uniform posterior distribution that ensures perfect anonymity.

2.6 Blanket Decomposition and Reduction

In contrast to the explicit formulas derived in Section 2.5, the posteriors of K and Y_1 lack closed-form expressions when user distributions P_i are heterogeneous. Nevertheless, for any privacy metric satisfying the data processing inequality (including mutual information, TV-information, and QIF measures), the analysis of the heterogeneous shuffle-only setting can be reduced to a tractable homogeneous case via blanket decomposition.

In the *shuffle-only* model, this reduction requires the inputs X_i to be *mutually independent* to ensure the existence of a well-defined post-processing function. In contrast, the *shuffle-DP* model (Section 6) circumvents this requirement by providing $\mathbf{X}_{-1}$ as side information to the post-processing step.

In the following, we introduce the blanket decomposition and reduction for the shuffle-only model; the corresponding techniques for the shuffle DP model are detailed in Section 6.

Definition 7 (Blanket Decomposition [BBGN19]). *Let $\gamma := \sum_{y \in \mathbb{Y}} \inf_{i \in [2, n]} P_i(y)$. The blanket distribution Q^B associated with $(P_2, \dots, P_n)$ is*

$$Q^B(y) = \frac{1}{\gamma} \inf_{i \in [2, n]} P_i(y).$$

Each P_i then admits the decomposition

$$P_i = \gamma Q^B + (1 - \gamma) \text{LO}_i,$$

where the left-over distribution LO_i is given by

$$\text{LO}_i(y) = \frac{P_i(y) - \gamma Q^B(y)}{1 - \gamma}, \quad \forall y \in \mathbb{Y}.$$

Operationally, sampling from P_i is equivalent to drawing from Q^B with probability γ , and from LO_i otherwise.

For analytical tractability, we utilize the *generalized blanket distribution* $\bar{Q}^B$ [SCW26b], defined on the augmented space $\bar{\mathbb{Y}} = \mathbb{Y} \cup \{\perp\}$:

$$\bar{Q}^B(y) = \begin{cases} \inf_{i \in [2, n]} P_i(y), & y \in \mathbb{Y}, \\ 1 - \gamma, & y = \perp. \end{cases}$$

Theorem 8 (Blanket Reduction [SCW26b]). *Let $\mathbf{Z}^r = \mathcal{S}(\{Y_1\} \cup \{Y'_i : i = 2, \dots, n\})$, where $Y'_i \stackrel{i.i.d.}{\sim} \bar{Q}^B$ and $\mathcal{S}$ applies a uniformly random permutation. Then there exists a post-processing function f^B such that*

$$f^B(\mathbf{Z}^r, \{\text{LO}_i\}_{i=2}^n) \stackrel{d}{=} \mathbf{Z},$$

i.e., the distribution of $\mathbf{Z}$ is obtained by post-processing $\mathbf{Z}^r$ using $\{\text{LO}_i\}_{i=2}^n$.

Operationally (see Algorithm 1), given $\mathbf{Z}^r$ containing c occurrences of $\perp$, the post-processing f^B selects c users uniformly at random and replaces those $\perp$ symbols with independent samples from the corresponding left-over distributions LO_i . Full proofs and procedural details appear in [SCW26b].

Theorem 9. *The mutual information satisfies $I(X_1; \mathbf{Z}) \leq I(X_1; \mathbf{Z}^r)$.*

Proof. By Theorem 8, given the set $\{\text{LO}_i\}_{i=2}^n$, we have the Markov chain $X_1 \rightarrow \mathbf{Z}^r \rightarrow \mathbf{Z}$. The result follows immediately from the data-processing inequality. $\square$

This reduction is particularly effective when the P_i are similar. If a particular P_i deviates substantially from the rest, one may exclude that user from the analysis: the output $\mathbf{Z}$ without that user's contribution can be post-processed to reconstruct the full $\mathbf{Z}$ by re-inserting that user's message.

Example 2.1. Suppose $P_1 = (0.4, 0.6)$, $P_2 = (0.3, 0.7)$, $P_3 = (0.5, 0.5)$, and $P_4 = (1.0, 0)$. User 4's distribution differs markedly from the others, so we may exclude it. The generalized blanket distribution of P_2 and P_3 is $\bar{Q}^B = (0.3, 0.5, 0.2)$, and

$$I(X_1; \mathbf{Z}) \leq I(X_1; \mathbf{Z}^r),$$

where $\mathbf{Z}^r = \mathcal{S}(\{Y_1\} \cup \{Y'_i : i = 2, 3\})$ and $Y'_i \stackrel{i.i.d.}{\sim} \bar{Q}^B$.

3 Optimal Blanket Noise under Mutual Information

This section presents a rigorous analysis of the mutual information leakages, $I(K; \mathbf{Z})$ and $I(Y_1; \mathbf{Z})$, within the canonical shuffle-only model. While classical results exist for the case where $P = Q$ (as reviewed in Section 1.6), we provide a foundational proof in Section 3.1 that serves as a basis for our generalization to the $Q \neq P$ regime in Section 3.2.

Remark 1. In the general heterogeneous setting where user distributions $\{P_i\}$ vary, the input leakage $I(Y_1; \mathbf{Z})$ can be analyzed via the blanket decomposition introduced in Section 2.6. For the position leakage $I(K; \mathbf{Z})$, analytical bounds can be established by imposing specific structural conditions on the distribution family $\{P_i\}$. Since this analysis is intrinsically linked to the shuffle-DP framework, a comprehensive discussion is deferred to Section 6.1.

3.1 Canonical Setting with $P = Q$

In this section, we analyze the shuffle information leakage when all users' messages are drawn from the same distribution P , i.e., $P = Q$.

Property 1. *In the canonical setting of the shuffle-only model, if $P = Q$, then $I(K; \mathbf{Z}) = 0$.*

Proof. By Theorem 7, the posterior distribution of K conditioned on any realization $\mathbf{Z} = \mathbf{z}$ is the uniform distribution U_n over $[n]$. Since the posterior coincides with the uniform prior, the mutual information $I(K; \mathbf{Z}) = \mathbb{E}_{\mathbf{Z}}[D_{\text{KL}}(\text{Law}(K | \mathbf{Z}) || U_n)]$ is identically zero. $\square$

Theorem 10. *In the canonical shuffle-only setting with $P = Q$ and support size m , the information leakage satisfies:*

$$I(Y_1; \mathbf{Z}) = \frac{m-1}{2n} + O(n^{-3/2}), \quad \text{as } n \rightarrow \infty.$$

Proof. We start by expanding the mutual information as

$$I(Y_1; \mathbf{Z}) = H(Y_1) - H(Y_1 | \mathbf{Z}) = H(Y_1) - [H(Y_1 | \mathbf{Z}, K) + I(Y_1; K | \mathbf{Z})]. \quad (2)$$

Since conditioning on the shuffle index K makes the message Y_1 directly observable in $\mathbf{Z}$, we have $H(Y_1 | \mathbf{Z}, K) = 0$. Then, using the chain rule for mutual information, $I(Y_1; K | \mathbf{Z}) = I(K; Y_1, \mathbf{Z}) - I(K; \mathbf{Z})$. Substituting this into (2) yields

$$I(Y_1; \mathbf{Z}) = H(Y_1) - I(K; Y_1, \mathbf{Z}) + I(K; \mathbf{Z}). \quad (3)$$

By the previous theorem, $I(K; \mathbf{Z}) = 0$ when $P = Q$. Thus, the information leakage reduces to computing $I(K; Y_1, \mathbf{Z})$.

Characterizing $I(K; Y_1, \mathbf{Z})$. Given $Y_1 = y_i$, the number of occurrences of y_i in $\mathbf{Z}$ follows $1 + X$, where $X \sim \text{Bin}(n-1, p_i)$. Since the posterior of K is uniform over all indices matching y_i , we have $H(K | \mathbf{Z}, Y_1 = y_i) = \mathbb{E}[\log(1 + X)]$. Following the combinatorial identity $\frac{1}{j+1} \binom{n-1}{j} = \frac{1}{n} \binom{n}{j+1}$, we obtain:

$$\begin{aligned} I(K; Y_1, \mathbf{Z}) &= H(K) - H(K; Y_1, \mathbf{Z}) \\ &= \log n - \sum_{i=1}^m p_i \mathbb{E}_{X \sim \text{Bin}(n-1, p_i)} [\log(1 + X)] \\ &= \log n - \sum_{i=1}^m \mathbb{E}_{X \sim \text{Bin}(n, p_i)} \left[\frac{X}{n} \log X \right] \\ &= - \sum_{i=1}^m \mathbb{E}_{X \sim \text{Bin}(n, p_i)} \left[\frac{X}{n} \log \left(\frac{X}{n} \right) \right]. \end{aligned} \quad (4)$$

Asymptotic Expansion. Substituting (4) into (3) and letting $f(t) = t \log t$, we have:

$$I(Y_1; \mathbf{Z}) = \sum_{i=1}^m \left(\mathbb{E} \left[f \left(\frac{X}{n} \right) \right] - f(p_i) \right).$$

A second-order Taylor expansion of $f(t)$ around $t = p_i$ gives

$$f(t) = p_i \log p_i + (\log p_i + 1)(t - p_i) + \frac{1}{2p_i}(t - p_i)^2 + O((t - p_i)^3). \quad (5)$$

For $X \sim \text{Bin}(n, p_i)$, the known moments are

$$\mathbb{E} \left[\frac{X}{n} - p_i \right] = 0, \quad \mathbb{E} \left[\left(\frac{X}{n} - p_i \right)^2 \right] = \text{Var} \left(\frac{X}{n} \right) = \frac{p_i(1 - p_i)}{n}.$$

Due to the Rosenthal inequality (Theorem 18 in Appendix B) for random variables,

$$\mathbb{E} \left[\left| \frac{X}{n} - p_i \right|^3 \right] = O(n^{-3/2}).$$

Taking expectations in (5) thus yields

$$\mathbb{E}_{X \sim \text{Bin}(n, p_i)} \left[\frac{X}{n} \log \left(\frac{X}{n} \right) \right] = p_i \log p_i + \frac{1 - p_i}{2n} + O(n^{-3/2}). \quad (6)$$

Some subtle details are provided in Appendix B.

Summing over all $i \in [m]$ yields the final result $\frac{m-1}{2n} + O(n^{-3/2})$. $\square$

Remark 2. When the Y_i are not mutually independent, the leakage $I(Y_1; \mathbf{Z})$ can be large even if all marginals are identical. For example, under full coherence $Y_1 = Y_2 = \dots = Y_n$ (each with marginal distribution P), the shuffle model outputs $\mathbf{Z}$ as n copies of the same symbol, thereby revealing Y_1 exactly. Consequently,

$$I(Y_1; \mathbf{Z}) = H(Y_1).$$

3.2 Canonical Setting with $P \ll Q$

Having treated the special case $P = Q$ as a warm-up, we now analyze the canonical setting under a general pair of distributions (P, Q) .

3.2.1 Analysis of Position Information Leakage

Proof of Theorem 1. Recall that K is uniform on $[n]$ and, conditional on $\mathbf{Z} = \mathbf{z} = (z_1, z_2, \dots, z_n)$,

$$\Pr[K = k \mid \mathbf{Z} = \mathbf{z}] = \frac{w(z_k)}{\sum_{i=1}^n w(z_i)}, \quad w(y) := \frac{P(y)}{Q(y)}. \quad (7)$$

Hence

$$\begin{aligned} I(K; \mathbf{Z}) &= \mathbb{E}_{\mathbf{Z}}[D_{\text{KL}}(\text{Law}(K \mid \mathbf{Z}) \parallel U_n)] \\ &= \mathbb{E}_{\mathbf{Z}} \left[\sum_{k=1}^n \Pr(K = k \mid \mathbf{Z}) \log(n \Pr(K = k \mid \mathbf{Z})) \right] \\ &= \log n + \mathbb{E}_{\mathbf{Z}} \left[\sum_{k=1}^n \frac{w(Z_k)}{S} \log \frac{w(Z_k)}{S} \right] \\ &= \log n + \mathbb{E}_{\mathbf{Z}} \left[\frac{1}{S} \sum_{k=1}^n w(Z_k) \log w(Z_k) \right] - \mathbb{E}_{\mathbf{Z}}[\log S], \end{aligned} \quad (8)$$

where

$$S := \sum_{k=1}^n w(Z_k), \quad \mathbf{Z} = (Z_1, \dots, Z_n).$$

By construction, among the n summands in S there is exactly one draw $W^{(P)} := w(Y_1)$ with $Y_1 \sim P$; the remaining $n - 1$ are i.i.d. $W_i^{(Q)} := w(Y_i)$ with $Y_i \sim Q$. We analyze the two expectations in (8) separately.

The “ratio” term $\mathbb{E}[\frac{1}{S} \sum w \log w]$. Let

$$T := \sum_{k=1}^n w(Z_k) \log w(Z_k) = W^{(P)} \log W^{(P)} + \sum_{i=2}^n W_i^{(Q)} \log W_i^{(Q)}.$$

An important property is that $\mathbb{E}_Q W = \sum_y Q(y) \cdot \frac{P(y)}{Q(y)} = 1$.

Define

$$\begin{aligned} D &:= \mathbb{E}_Q[W \log W] = \sum_y Q(y) \frac{P(y)}{Q(y)} \log \frac{P(y)}{Q(y)} = D_{\text{KL}}(P \parallel Q), \\ \chi^2 &:= \text{Var}_Q(W) = \sum_y Q(y) \left(\frac{P(y)}{Q(y)} - 1 \right)^2 = \chi^2(P \parallel Q), \end{aligned}$$

$$B := \mathbb{E}_Q[W^2 \log W].$$

Then

$$\mathbb{E}_P[W] = \mathbb{E}_Q[W^2] = \sum_y \frac{P(y)^2}{Q(y)} = 1 + \chi^2, \quad \mathbb{E}_P[W \log W] = \mathbb{E}_Q[W^2 \log W] = B. \quad (9)$$

With $\mu_S := \mathbb{E}[S] = n + \chi^2$ and $\mu_T := \mathbb{E}[T] = nD + (B - D)$, a second-order delta-method (Lemma 4 in Appendix C) gives

$$\mathbb{E}\left[\frac{T}{S}\right] = \frac{\mu_T}{\mu_S} - \frac{\text{Cov}(T, S)}{\mu_S^2} + \frac{\mu_T \text{Var}(S)}{\mu_S^3} + O(n^{-3/2}). \quad (10)$$

Using

$$\begin{aligned} \text{Var}(S) &= (n-1)\text{Var}_Q(W) + \text{Var}_P(W) = n\chi^2 + O(1), \\ \text{Cov}(T, S) &= (n-1)\text{Cov}_Q(W, W \log W) + O(1) \\ &= (n-1)(\mathbb{E}_Q[W^2 \log W] - \mathbb{E}_Q[W] \cdot \mathbb{E}_Q[W \log W]) + O(1) \\ &= (n-1)(B - D) + O(1), \end{aligned}$$

substituting into (10) yields

$$\begin{aligned} \mathbb{E}\left[\frac{T}{S}\right] &= \frac{nD + (B - D)}{n + \chi^2} - \frac{\text{Cov}(T, S)}{\mu_S^2} + \frac{\mu_T \text{Var}(S)}{\mu_S^3} + O(n^{-3/2}) \\ &= D + \frac{1}{n}((B - D) - D\chi^2) - \frac{1}{n}(B - D) + \frac{1}{n}D\chi^2 + O(n^{-3/2}) \\ &= D + O(n^{-3/2}). \end{aligned} \quad (11)$$

Thus the ratio term contributes $D_{\text{KL}}(P\|Q)$ up to $O(n^{-3/2})$.

The term $\mathbb{E}[\log S]$. Write $S = n + U$ with $\mathbb{E}[U] = \chi^2$ and $\text{Var}(U) = \text{Var}(S) = n\chi^2 + O(1)$. Expanding

$$\log(n + U) = \log n + \frac{U}{n} - \frac{U^2}{2n^2} + O\left(\frac{|U|^3}{n^3}\right),$$

and using

$$\begin{aligned} \mathbb{E}[U^2] &= \text{Var}[U] + (\mathbb{E}[U])^2 = n\chi^2 + O(1), \\ \mathbb{E}[|U|^3] &= O\left((\text{Var}[U])^{3/2} + (n-1)\mathbb{E}_Q|W-1|^3 + \mathbb{E}_P|W-1|^3 + |\mathbb{E}[U]|^3\right) = O(n^{3/2}), \end{aligned}$$

(by Rosenthal inequality, Theorem 18), we obtain

$$\mathbb{E}[\log S] = \log n + \frac{\chi^2}{n} - \frac{n\chi^2 + O(1)}{2n^2} + O(n^{-3/2}) = \log n + \frac{\chi^2}{2n} + O(n^{-3/2}). \quad (12)$$

Conclusion. Combining (8), (11), and (12),

$$I(K; \mathbf{Z}) = \log n + D_{\text{KL}}(P\|Q) - \left(\log n + \frac{\chi^2(P\|Q)}{2n}\right) + O(n^{-3/2}),$$

which simplifies to the claimed expansion. $\square$

3.2.2 Analysis of Message Information Leakage

Proof of Theorem 2. We analyze the mutual information using the same decomposition as in Eq. (3):

$$I(Y_1; \mathbf{Z}) = H(Y_1) - I(K; Y_1, \mathbf{Z}) + I(K; \mathbf{Z}).$$

Since $I(K; \mathbf{Z})$ has already been derived in Theorem 1 as

$$I(K; \mathbf{Z}) = D_{\text{KL}}(P\|Q) - \frac{\chi^2(P\|Q)}{2n} + O(n^{-3/2}),$$

it remains to compute the term $H(Y_1) - I(K; Y_1, \mathbf{Z})$.

Expanding $H(Y_1) - I(K; Y_1, \mathbf{Z})$. Similar to the derivation of Theorem 10, we have

$$H(Y_1) - I(K; Y_1, \mathbf{Z}) = \sum_{i=1}^m p_i \mathbb{E}_{X \sim \text{Bin}(n-1, q_i)} \left[\frac{X+1}{n} \log \left(\frac{X+1}{n} \right) \right] - p_i \log p_i. \quad (13)$$

Using similar techniques (the combinatorial identity $\frac{1}{m} \binom{n-1}{m-1} = \frac{1}{n} \binom{n}{m}$), we have

$$H(Y_1) - I(K; Y_1, \mathbf{Z}) = \sum_{i=1}^m \frac{p_i}{q_i} \mathbb{E}_{X \sim \text{Bin}(n, q_i)} \left[\frac{X}{n} \log \left(\frac{X}{n} \right) \right] - p_i \log p_i. \quad (14)$$

Let $f(t) = t \log t$. Expanding around $t = q_i$, we have

$$f(t) = q_i \log q_i + (\log q_i + 1)(t - q_i) + \frac{1}{2q_i}(t - q_i)^2 + O((t - q_i)^3).$$

For $X \sim \text{Bin}(n, q_i)$, the moments are

$$\mathbb{E} \left[\frac{X}{n} \right] = q_i, \quad \text{Var} \left(\frac{X}{n} \right) = \frac{q_i(1 - q_i)}{n}.$$

Due to the Rosenthal inequality, $\mathbb{E} \left[\left| \frac{X}{n} - q_i \right|^3 \right] = O(n^{-3/2})$.

Taking expectations in the expansion gives

$$\mathbb{E}_{X \sim \text{Bin}(n, q_i)} \left[\frac{X}{n} \log \left(\frac{X}{n} \right) \right] = q_i \log q_i + \frac{1 - q_i}{2n} + O(n^{-3/2}). \quad (15)$$

Plugging (15) into (14) yields

$$H(Y_1) - I(K; Y_1, \mathbf{Z}) = \sum_i p_i \log \left(\frac{q_i}{p_i} \right) + \frac{p_i(1 - q_i)}{2q_i n} + O(n^{-3/2}). \quad (16)$$

The first summation equals $-D_{\text{KL}}(P\|Q)$, and since

$$\sum_i \frac{p_i(1 - q_i)}{q_i} = -1 + \sum_i \frac{p_i}{q_i},$$

we can rewrite (16) as

$$H(Y_1) - I(K; Y_1, \mathbf{Z}) = -D_{\text{KL}}(P\|Q) + \frac{-1 + \sum_i \frac{p_i}{q_i}}{2n} + O(n^{-3/2}). \quad (17)$$

Finally, substituting (17) and $I(K; \mathbf{Z})$ into the decomposition gives

$$\begin{aligned} I(Y_1; \mathbf{Z}) &= H(Y_1) - I(K; Y_1, \mathbf{Z}) + I(K; \mathbf{Z}) \\ &= \frac{1}{2n} \sum_i \frac{p_i - p_i^2}{q_i} + O(n^{-3/2}). \end{aligned} \quad (18)$$

This completes the proof. $\square$

3.2.3 Optimal Blanket under Mutual Information

Proof of Theorem 3. According to Theorem 2, the leading term of $I(Y_1; \mathbf{Z})$ is $\frac{1}{2n} \sum_i \frac{p_i(1-p_i)}{q_i}$. Thus, in asymptotic sense, minimizing $I(Y_1; \mathbf{Z})$ over Q reduces to

$$\min_{\{q_i\}} \sum_i \frac{a_i}{q_i} \quad \text{subject to} \quad q_i \geq 0, \sum_i q_i = 1, \quad \text{where } a_i := p_i(1-p_i).$$

By Cauchy–Schwarz inequality,

$$\sum_i \frac{a_i}{q_i} = \sum_i \frac{(\sqrt{a_i})^2}{q_i} \geq \frac{(\sum_i \sqrt{a_i})^2}{\sum_i q_i} = \left(\sum_i \sqrt{a_i} \right)^2,$$

with equality if and only if $\frac{a_i}{q_i} = \lambda q_i$ for some $\lambda > 0$, i.e., $q_i \propto \sqrt{a_i} = \sqrt{p_i(1-p_i)}$. $\square$

Corollary 1. *If $\text{Supp}(P)$ has size 2, then the asymptotically optimal Q is the uniform distribution on $\text{Supp}(P)$.*

Proof. Let the two support points have probabilities p and $1-p$. Then $\sqrt{p(1-p)} = \sqrt{(1-p)p}$ for both points, so the rule $q_i \propto \sqrt{p_i(1-p_i)}$ assigns equal mass to each support point. $\square$

4 Optimal Blanket Noise under Total-Variation-Information

4.1 Analysis of Position Information Leakage

Proof of Theorem 4. Let $w(x) = \frac{P(x)}{Q(x)}$ denote the likelihood ratio between P and Q .

Note that $\mathbb{E}_{A \sim Q}[w(A)] = \int \frac{P(x)}{Q(x)} dQ(x) = 1$. Let $A_1, A_2, \dots$ be i.i.d. samples drawn from Q , and define the partial sum $S_{n-1} = \sum_{i=1}^{n-1} w(A_i)$.

By the definition of TV-information in the canonical shuffle-only setting, we have:

$$I_{\text{TV}}(K; \mathbf{Z}) = \frac{1}{2} \mathbb{E}_{\mathbf{Z}} \left[\sum_{i=1}^n \left| \frac{w(z_i)}{\sum_{j=1}^n w(z_j)} - \frac{1}{n} \right| \right].$$

Recall that the density ratio between the joint distribution $\Pr[\mathbf{Z}]$ and the product measure Q^n is $\frac{1}{n} \sum_{j=1}^n w(z_j)$. Performing a change of measure to the i.i.d. space Q^n , we obtain:

$$\begin{aligned} I_{\text{TV}}(K; \mathbf{Z}) &= \frac{1}{2} \mathbb{E}_{Q^n} \left[\frac{\sum_{j=1}^n w(z_j)}{n} \sum_{i=1}^n \left| \frac{w(z_i)}{\sum_{j=1}^n w(z_j)} - \frac{1}{n} \right| \right] \\ &= \frac{1}{2} \mathbb{E}_{Q^n} \left[\frac{1}{n} \sum_{i=1}^n \left| w(z_i) - \frac{1}{n} \sum_{j=1}^n w(z_j) \right| \right]. \end{aligned}$$

Due to the symmetry of the i.i.d. samples under Q^n , the expectation of the sum is n times the expectation of a single term. Let z_1 represent the first sample and $S_{n-1} = \sum_{j=2}^n W(z_j)$. Then:

$$\begin{aligned} I_{\text{TV}}(K; \mathbf{Z}) &= \frac{1}{2} \mathbb{E}_{z_1 \sim Q, S_{n-1}} \left[\left| w(z_1) - \frac{w(z_1) + S_{n-1}}{n} \right| \right] \\ &= \frac{1}{2} \left(\frac{n-1}{n} \right) \mathbb{E}_{z_1 \sim Q, S_{n-1}} \left[\left| w(z_1) - \frac{S_{n-1}}{n-1} \right| \right]. \end{aligned}$$

By the Strong Law of Large Numbers (SLLN), as $n \rightarrow \infty$, the sample average $\frac{S_{n-1}}{n-1}$ converges almost surely to $\mathbb{E}_Q[w(A)] = 1$. Combined with the Dominated Convergence Theorem (as the expression is bounded in L_1), we have:

$$\begin{aligned} \lim_{n \rightarrow \infty} I_{\text{TV}}(K; \mathbf{Z}) &= \frac{1}{2} \mathbb{E}_{z_1 \sim Q} [|w(z_1) - 1|] \\ &= \frac{1}{2} \sum_{z_1 \in \text{supp}(Q)} |P(z_1) - Q(z_1)| = D_{\text{TV}}(P, Q). \end{aligned}$$

This completes the proof. $\square$

4.2 Analysis of Message Information Leakage

Proof of Theorem 5. Let Q^n denote the reference measure where $\mathbf{Z}$ consists of n i.i.d. samples from Q . Under the canonical shuffle-only model, the likelihood ratio between the conditional distribution and the reference measure is $\frac{d\mathbb{P}(\mathbf{z} | Y_1=i)}{dQ^n(\mathbf{z})} = \frac{n_i}{nq_i}$. By linearity, the marginal ratio is $\frac{d\mathbb{P}(\mathbf{z})}{dQ^n(\mathbf{z})} = \sum_j p_j \frac{n_j}{nq_j}$.

Starting from the definition of TV-information and performing a change of measure to Q^n :

$$\begin{aligned} I_{\text{TV}}(Y_1; \mathbf{Z}) &= \sum_i p_i D_{\text{TV}}(\mathbb{P}_{\mathbf{Z}|i}, \mathbb{P}_{\mathbf{Z}}) \\ &= \frac{1}{2} \sum_i p_i \int_{\mathcal{Z}} \left| \frac{d\mathbb{P}(\mathbf{z} | i)}{dQ^n(\mathbf{z})} - \frac{d\mathbb{P}(\mathbf{z})}{dQ^n(\mathbf{z})} \right| dQ^n(\mathbf{z}) \\ &= \frac{1}{2} \sum_i p_i \mathbb{E}_{\tilde{\mathbf{n}} \sim \text{Multi}(n, Q)} \left[\left| \frac{n_i}{nq_i} - \sum_j p_j \frac{n_j}{nq_j} \right| \right]. \end{aligned}$$

The result follows immediately. $\square$

4.3 Optimal Blanket for Binary Source under TV-Information

Proof of Theorem 6. Without loss of generality, assume the message space is $\mathcal{M} = \{0, 1\}$. Let the target distribution be $P = (1-p, p)$ and the blanket noise distribution be $Q = (1-q, q)$. Let $\mathbf{Z}$ be the shuffled output consisting of one sample from P and $n-1$ i.i.d. samples from Q . Let n_1 and $n_0 = n - n_1$ denote the frequencies of 1 and 0 in $\mathbf{Z}$, respectively.

$$I_{\text{TV}}(Y_1; \mathbf{Z}) = \mathbb{E}_{\mathbf{Z}} [D_{\text{TV}}(P, \text{Law}(Y_1 | \mathbf{Z}))] = \mathbb{E}_{\mathbf{Z}} [|p - \mathbb{P}(Y_1 = 1 | \mathbf{Z})|].$$

By Bayes' rule, the posterior probability is given by:

$$\mathbb{P}(Y_1 = 1 | \mathbf{Z}) = \frac{p \cdot \mathbb{P}(\mathbf{Z} | Y_1 = 1)}{p \cdot \mathbb{P}(\mathbf{Z} | Y_1 = 1) + (1-p) \cdot \mathbb{P}(\mathbf{Z} | Y_1 = 0)} = \frac{p \frac{n_1}{nq}}{p \frac{n_1}{nq} + (1-p) \frac{n_0}{n(1-q)}},$$

where the last equality follows from the fact that given $Y_1 = y$, the remaining $n - 1$ messages follow $\text{Bin}(n - 1, q)$. Substituting this into the TV expression and letting $\mathbf{Z}' \sim \text{Bin}(n, q)$, we have:

$$\begin{aligned} I_{TV}(Y_1; \mathbf{Z}) &= \mathbb{E}_{\mathbf{Z}'} \left[\left| p - \frac{p \frac{n_1}{q}}{p \frac{n_1}{q} + (1-p) \frac{n_0}{1-q}} \right| \cdot \frac{\mathbb{P}(\mathbf{Z})}{\mathbb{P}(\mathbf{Z}')} \right] \\ &= \mathbb{E}_{\mathbf{Z}'} \left| p \left(p \frac{n_1}{nq} + (1-p) \frac{n_0}{n(1-q)} \right) - p \frac{n_1}{nq} \right| \\ &= \frac{p(1-p)}{nq(1-q)} \mathbb{E}_{n_1 \sim \text{Bin}(n, q)} [|n_1 - nq|]. \end{aligned}$$

The proof of this theorem is implied by Lemma 3. □

Lemma 3. *Let $n \in \mathbb{N}$ and $q \in (0, 1)$. Define the function*

$$\phi(q) = \frac{1}{q(1-q)} \mathbb{E}_{X \sim \text{Bin}(n, q)} [|X - nq|].$$

The global minimum of $\phi(q)$ is attained at $q = \frac{\lfloor n/2 \rfloor}{n}$ and $q = \frac{\lceil n/2 \rceil}{n}$.

Proof of Lemma 3. The Mean Absolute Deviation of a binomial random variable $X \sim \text{Bin}(n, q)$ is given by the following identity (originally attributed to De Moivre [DZ91, BK13]):

$$\phi(q) = \frac{2 \binom{n-1}{k} q^{k+1} (1-q)^{n-k}}{q(1-q)} = 2 \binom{n-1}{k} q^k (1-q)^{n-k-1}, \text{ where } k = \lfloor nq \rfloor$$

Since $k = \lfloor nq \rfloor$ is constant on the intervals $I_k = [\frac{k}{n}, \frac{k+1}{n})$ for $k \in \{0, 1, \dots, n-1\}$, we analyze $\phi(q)$ as a sequence of functions $f_k(q) = 2 \binom{n-1}{k} q^k (1-q)^{n-k-1}$ defined on each I_k .

To find the extremum of $f_k(q)$ within I_k , we take the derivative with respect to q :

$$\begin{aligned} f'_k(q) &= 2 \binom{n-1}{k} \frac{d}{dq} (q^k (1-q)^{n-k-1}) \\ &= 2 \binom{n-1}{k} [k q^{k-1} (1-q)^{n-k-1} - (n-k-1) q^k (1-q)^{n-k-2}] \\ &= 2 \binom{n-1}{k} q^{k-1} (1-q)^{n-k-2} [k(1-q) - (n-k-1)q] \\ &= 2 \binom{n-1}{k} q^{k-1} (1-q)^{n-k-2} [k - (n-1)q]. \end{aligned}$$

Setting $f'_k(q) = 0$ yields the critical point $q_k^* = \frac{k}{n-1}$. Note that for $k \in \{1, \dots, n-2\}$, the point q_k^* always lies within the interval I_k since:

$$\frac{k}{n} \leq \frac{k}{n-1} \leq \frac{k+1}{n}$$

holds for all $n \geq 2$. Since $f_k(q)$ is a unimodal function that increases for $q < q_k^*$ and decreases for $q > q_k^*$, it reaches a *local maximum* at q_k^* .

Because $\phi(q)$ is continuous on $(0, 1)$ and unimodal within each interval I_k , the **local minima** of $\phi(q)$ must occur at the boundary points of these intervals, i.e., at $q \in \{\frac{k}{n}\}_{k=1}^{n-1}$.

Let $a_k = \phi(\frac{k}{n})$. We wish to find k that minimizes:

$$a_k = 2 \binom{n-1}{k} \left(\frac{k}{n}\right)^k \left(\frac{n-k}{n}\right)^{n-k-1}.$$

Consider the ratio of consecutive terms:

$$\begin{aligned} \frac{a_{k+1}}{a_k} &= \frac{\binom{n-1}{k+1} \left(\frac{k+1}{n}\right)^{k+1} \left(\frac{n-k-1}{n}\right)^{n-k-2}}{\binom{n-1}{k} \left(\frac{k}{n}\right)^k \left(\frac{n-k}{n}\right)^{n-k-1}} \\ &= \frac{n-k-1}{k+1} \cdot \frac{(k+1)^{k+1}}{k^k} \cdot \frac{(n-k-1)^{n-k-2}}{(n-k)^{n-k-1}} \\ &= \frac{(k+1)^k}{k^k} \cdot \frac{(n-k-1)^{n-k-1}}{(n-k)^{n-k-1}} \\ &= \left(\frac{k+1}{k}\right)^k \cdot \left(\frac{n-k-1}{n-k}\right)^{n-k-1}. \end{aligned}$$

Let $g(x) = (1 + 1/x)^x$. It is well known that $g(x)$ is strictly increasing and converges to e as $x \rightarrow \infty$. The ratio can be written as:

$$\frac{a_{k+1}}{a_k} = \frac{g(k)}{g(n-k-1)}.$$

- If $k < n-k-1$ (i.e., $k < \frac{n-1}{2}$), then $g(k) < g(n-k-1)$, which implies $\frac{a_{k+1}}{a_k} < 1$. Thus, the sequence a_k is strictly decreasing.
- If $k > n-k-1$ (i.e., $k > \frac{n-1}{2}$), then $g(k) > g(n-k-1)$, which implies $\frac{a_{k+1}}{a_k} > 1$. Thus, the sequence a_k is strictly increasing.
- If n is odd and $k = \frac{n-1}{2}$, then $k = n-k-1$, implying $\frac{a_{k+1}}{a_k} = 1$, so $a_{(n-1)/2} = a_{(n+1)/2}$.

Conclusion. The sequence a_k reaches its minimum when k is as close to $n/2$ as possible. Specifically:

- If n is even, the minimum is at $k = n/2$, corresponding to $q = 1/2$.
- If n is odd, the minimum is at $k = (n-1)/2$ and $k = (n+1)/2$, corresponding to $q = \frac{n-1}{2n}$ and $q = \frac{n+1}{2n}$.

This completes the proof. $\square$

5 Optimal Blanket Noise under Quantitative Information Flow

5.1 Analysis of Message Recovery Advantage

Theorem 11. Consider the canonical shuffle-only model setting where the target user's input $Y_1 \sim P$ and $n-1$ blanket messages i.i.d. according to Q . Assume $\text{Supp}(P) \subseteq \text{Supp}(Q) \subseteq [m] = \{1, \dots, m\}$. The expected message recovery advantage, is given by:

$$\text{Adv}_{\text{MR}}(Y_1; \mathbf{Z}) = \mathbb{E}_{\vec{n} \sim \text{Multi}(n, Q)} \left[\max_{i \in [m]} \frac{p_i n_i}{q_i n} \right] - \max_{i \in [m]} p_i, \quad (19)$$

where $\vec{n} = (n_1, \dots, n_m)$ denotes the vector of counts of n i.i.d. samples from Q .

Proof. Let Q^n denote the reference measure of n i.i.d. samples from Q . Recall that the density ratio between the marginal distribution $\mathbb{P}_{\mathbf{Z}}$ and the reference measure is $\frac{d\mathbb{P}(\mathbf{z})}{dQ^n(\mathbf{z})} = \sum_{j=1}^m p_j \frac{n_j}{nq_j}$, and the posterior probability of $Y_1 = i$ given $\mathbf{z}$ is:

$$\mathbb{P}(Y_1 = i | \mathbf{z}) = \frac{p_i \frac{n_i}{nq_i}}{\sum_{j=1}^m p_j \frac{n_j}{nq_j}}.$$

By performing a change of measure from $\mathbb{P}_{\mathbf{Z}}$ to Q^n , the expected maximum posterior probability can be expressed as:

$$\begin{aligned} \mathbb{E}_{\mathbf{Z}} \left[\max_{i \in [m]} \mathbb{P}(Y_1 = i | \mathbf{z}) \right] &= \int_{\mathbf{Z}} \max_{i \in [m]} \left(\frac{p_i \frac{n_i}{nq_i}}{\sum_{j=1}^m p_j \frac{n_j}{nq_j}} \right) \frac{d\mathbb{P}(\mathbf{z})}{dQ^n(\mathbf{z})} dQ^n(\mathbf{z}) \\ &= \mathbb{E}_{\tilde{\mathbf{n}} \sim \text{Multi}(n, Q)} \left[\max_{i \in [m]} \left(\frac{p_i \frac{n_i}{nq_i}}{\sum_{j=1}^m p_j \frac{n_j}{nq_j}} \right) \left(\sum_{j=1}^m p_j \frac{n_j}{nq_j} \right) \right] \\ &= \mathbb{E}_{\tilde{\mathbf{n}} \sim \text{Multi}(n, Q)} \left[\max_{i \in [m]} \frac{p_i n_i}{nq_i} \right]. \end{aligned}$$

The result follows. $\square$

5.2 Analysis of Expected Posterior Variance

Theorem 12. Consider the canonical shuffle-only model where the target user's input $Y_1 \sim P$ and $n - 1$ blanket messages are i.i.d. samples from Q , with $\text{Supp}(P) \subseteq \text{Supp}(Q) \subseteq \mathbb{R}$. Define the likelihood ratio (weight function) $w(y) := \frac{P(y)}{Q(y)}$. Let $A_1, A_2, \dots, A_n \stackrel{i.i.d.}{\sim} Q$, and define the partial sums of weights as $S_k = \sum_{i=1}^k w(A_i)$.

The expected posterior variance $\mathbb{E}_{\mathbf{Z}}[\text{Var}(Y_1 | \mathbf{Z})]$ is given by:

$$\mathbb{E}_{\mathbf{Z}}[\text{Var}(Y_1 | \mathbf{Z})] = \mathbb{E}_{y \sim P}[y^2] - \mathbb{E}_{y \sim P, S_{n-1}} \left[\frac{w(y)y^2}{w(y) + S_{n-1}} \right] - (n-1) \mathbb{E}_{y_1, y_2 \sim P, S_{n-2}} \left[\frac{y_1 y_2}{w(y_1) + w(y_2) + S_{n-2}} \right].$$

Proof of Theorem 12. Let $\mathbf{Z} = (z_1, \dots, z_n)$ be the observed shuffled multiset. In the canonical setting, the posterior distribution of the target index $K \in \{1, \dots, n\}$ is $\pi_i = \mathbb{P}(Y_1 = z_i | \mathbf{Z}) = \frac{w(z_i)}{\beta}$, where $\beta = \sum_{j=1}^n w(z_j)$. The posterior variance is defined as:

$$\text{Var}(Y_1 | \mathbf{Z}) = \mathbb{E}_{Y_1 | \mathbf{Z}}[Y_1^2] - (\mathbb{E}_{Y_1 | \mathbf{Z}}[Y_1])^2 = \sum_{i=1}^n \pi_i z_i^2 - \left(\sum_{i=1}^n \pi_i z_i \right)^2.$$

Let $\mathbf{Z}' = (z_1, \dots, z_n)$ be a sequence of n i.i.d. samples from Q . The marginal distribution of the shuffled output $\mathbf{Z}$ relates to the i.i.d. measure Q^n via the density ratio $\frac{d\mathbb{P}_{\mathbf{Z}}}{dQ^n} = \frac{\beta}{n}$. Thus, for any functional $f(\mathbf{Z})$, we have $\mathbb{E}_{\mathbf{Z}}[f(\mathbf{Z})] = \mathbb{E}_{Q^n} \left[\frac{\beta}{n} f(\mathbf{Z}') \right]$. Applying this to the components of the variance:

$$\mathbb{E}_{\mathbf{Z}} \left[\sum_{i=1}^n \pi_i z_i^2 \right] = \mathbb{E}_{Q^n} \left[\frac{\beta}{n} \sum_{i=1}^n \frac{w(z_i)}{\beta} z_i^2 \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{Q^n}[w(z_i) z_i^2].$$

By the i.i.d. property and the definition of $w(z)$, $\mathbb{E}_{z \sim Q}[w(z)z^2] = \int Q(z) \frac{P(z)}{Q(z)} z^2 dz = \mathbb{E}_{y \sim P}[y^2]$. Thus, the first term simplifies to $\mathbb{E}_{y \sim P}[y^2]$.

Expanding the square $(\sum \pi_i z_i)^2 = \sum \pi_i^2 z_i^2 + \sum_{i \neq j} \pi_i \pi_j z_i z_j$, we compute its expectation:

$$\begin{aligned} \mathbb{E}_{\mathbf{Z}} [(\mathbb{E}_{Y_1|\mathbf{Z}}[Y_1])^2] &= \mathbb{E}_{Q^n} \left[\frac{\beta}{n} \left(\sum_{i=1}^n \frac{w(z_i)^2}{\beta^2} z_i^2 + \sum_{i \neq j} \frac{w(z_i)w(z_j)}{\beta^2} z_i z_j \right) \right] \\ &= \mathbb{E}_{Q^n} \left[\frac{1}{n\beta} \sum_{i=1}^n w(z_i)^2 z_i^2 \right] + \mathbb{E}_{Q^n} \left[\frac{1}{n\beta} \sum_{i \neq j} w(z_i)w(z_j) z_i z_j \right]. \end{aligned}$$

By symmetry, the n terms in the first sum are identical, as are the $n(n-1)$ terms in the second sum. Substituting $\beta = w(z_i) + S_{n-1}$ or $\beta = w(z_i) + w(z_j) + S_{n-2}$:

$$\begin{aligned} \text{Term 1} &= \mathbb{E}_{Q^n} \left[\frac{w(z_1)^2 z_1^2}{w(z_1) + S_{n-1}} \right] = \mathbb{E}_{z_1 \sim Q} \left[\mathbb{E}_{S_{n-1}} \left[\frac{w(z_1)^2 z_1^2}{w(z_1) + S_{n-1}} \right] \right] = \mathbb{E}_{y \sim P, S_{n-1}} \left[\frac{w(y)y^2}{w(y) + S_{n-1}} \right]. \\ \text{Term 2} &= (n-1) \mathbb{E}_{Q^n} \left[\frac{w(z_1)w(z_2)z_1 z_2}{w(z_1) + w(z_2) + S_{n-2}} \right] = (n-1) \mathbb{E}_{y_1, y_2 \sim P, S_{n-2}} \left[\frac{y_1 y_2}{w(y_1) + w(y_2) + S_{n-2}} \right]. \end{aligned}$$

Combining these yields the theorem. $\square$

6 Analysis of the Shuffle-DP Model

In this section, we analyze the mutual information in the shuffle DP model. Each user applies an ε_0 -local randomizer $\mathcal{R}$ to their input X_i , producing $Y_i = \mathcal{R}(X_i)$, $i = 1, \dots, n$, and the outputs are subsequently shuffled by a uniformly random permutation σ , yielding the shuffled view $\mathbf{Z} = (Y_{\sigma(i)})_{i=1}^n$. Let $\mathbf{X} = (X_1, X_2, X_3, \dots, X_n)$ and $\mathbf{X}_{-1} = (X_2, X_3, \dots, X_n)$.

6.1 Analysis of $I(K; \mathbf{Z})$

Theorem 13. *In the single-message shuffle-DP model, the shuffled output $\mathbf{Z}$ is $2\varepsilon_0$ -DP with respect to the shuffle index K , for any given $\mathbf{X}$. Consequently,*

$$I(K; \mathbf{Z} \mid \mathbf{X}) \leq 2\varepsilon_0.$$

Proof. Differential privacy with respect to K is determined by bounding the ratio $\Pr[\mathbf{Z} = \mathbf{z} \mid K = k, \mathbf{X} = \mathbf{x}] / \Pr[\mathbf{Z} = \mathbf{z} \mid K = k', \mathbf{X} = \mathbf{x}]$ for any two positions $k, k' \in [n]$ and any fixed $\mathbf{x}$.

Expansion of the conditional likelihood. Conditioned on $K = k$, the message in position k originates from user 1, while all other messages are outputs of the remaining users. Then

$$\Pr[\mathbf{Z} = \mathbf{z} \mid K = k, \mathbf{X} = \mathbf{x}] = \sum_{\sigma: \sigma(k)=1} \frac{1}{(n-1)!} \prod_{i=1}^n \mathcal{R}_{x_{\sigma(i)}}(z_i). \quad (20)$$

Similarly for $K = k'$:

$$\Pr[\mathbf{Z} = \mathbf{z} \mid K = k', \mathbf{X} = \mathbf{x}] = \sum_{\sigma: \sigma(k')=1} \frac{1}{(n-1)!} \prod_{i=1}^n \mathcal{R}_{x_{\sigma(i)}}(z_i). \quad (21)$$

Applying the LDP guarantee. Because $\mathcal{R}$ is ε_0 -locally differentially private, its output distributions satisfy, for all users $t \in [n] \setminus \{1\}$ and all z in the output domain,

$$e^{-\varepsilon_0} \leq \frac{\mathcal{R}_{x_1}(z)}{\mathcal{R}_{x_t}(z)} \leq e^{\varepsilon_0}.$$

Now fix k and k' . Each permutation σ contributing to Equation (20) can be mapped one-to-one to a permutation σ' in Equation (21) that swaps the roles of users 1 and $t = \sigma(k')$. This exchange changes at most two output terms in the product—those corresponding to z_k and $z_{k'}$ —each incurring at most an e^{ε_0} multiplicative change by the LDP guarantee. Hence, for every such pair of permutations,

$$\prod_{i=1}^n \mathcal{R}_{x_{\sigma(i)}}(z_i) \leq e^{2\varepsilon_0} \prod_{i=1}^n \mathcal{R}_{x_{\sigma'(i)}}(z_i).$$

Summing over all σ and using the bijection between σ and σ' gives

$$\Pr[\mathbf{Z} = z \mid K = k, \mathbf{X} = \mathbf{x}] \leq e^{2\varepsilon_0} \Pr[\mathbf{Z} = z \mid K = k', \mathbf{X} = \mathbf{x}].$$

That is, the shuffled output distribution is $2\varepsilon_0$ -DP with respect to the index K given $\mathbf{X}$.

Differential privacy and mutual information. It is known that if a random variable Z satisfies ε -LDP with respect to a discrete input K , then $I(K; Z) \leq \varepsilon$ [CY16]⁶. In our case, this yields

$$I(K; \mathbf{Z} \mid \mathbf{X}) \leq 2\varepsilon_0,$$

completing the proof. $\square$

Corollary 2. *In the single-message shuffle DP model, the information leakage about the position K is bounded by*

$$I(K; \mathbf{Z}) \leq 2\varepsilon_0.$$

Proof. Using the chain rule for mutual information,

$$I(K; \mathbf{Z} \mid \mathbf{X}) = I(K; \mathbf{Z}) + I(K; \mathbf{X} \mid \mathbf{Z}) - I(K; \mathbf{X}).$$

Since K is independent of $\mathbf{X}$, we have $I(K; \mathbf{X}) = 0$. Hence,

$$I(K; \mathbf{Z}) = I(K; \mathbf{Z} \mid \mathbf{X}) - I(K; \mathbf{X} \mid \mathbf{Z}) \leq I(K; \mathbf{Z} \mid \mathbf{X}) \leq 2\varepsilon_0,$$

because $I(K; \mathbf{X} \mid \mathbf{Z}) \geq 0$ and Theorem 13. $\square$

The same technique yields an information leakage bound for $I(K; \mathbf{Z})$ in the general shuffle-only setting:

Theorem 14. *In the general shuffle-only setting, suppose the distributions $\{P_i\}_{i=1}^n$ satisfy*

$$\forall y \in \mathbb{Y}, i, j \in [n], \quad \frac{P_i(y)}{P_j(y)} \leq e^{\varepsilon_0}.$$

Then

$$I(K; \mathbf{Z}) \leq 2\varepsilon_0.$$

Proof. Define a local randomizer $\mathcal{R}$ on $[n]$ by $\mathcal{R}(i) = P_i$. The shuffle-only model can then be viewed as a special case of the shuffle-DP setting with fixed inputs $\mathbf{X} = (1, 2, \dots, n)$. Applying Theorem 13 directly yields the claimed bound. $\square$

⁶There are more refined bounds like $\min\{\varepsilon^2, \varepsilon\}$ [CY16].

6.2 Analysis of $I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1})$

6.2.1 Blanket Decomposition and Reduction in Shuffle DP Model

Analogous to the approach in Section 2.6, the shuffle-DP setting can be reduced to a simpler and analytically tractable *canonical setting*. This reduction relies on the following decomposition of the local randomizer $\mathcal{R}$.

Definition 8 (Blanket decomposition of $\mathcal{R}$ [BBGN19]). *Let $\gamma = \sum_{y \in \mathbb{Y}} \inf_x \mathcal{R}_x(y)$. The blanket distribution Q^B of an ε_0 -DP mechanism $\mathcal{R}$ is defined by*

$$Q^B(y) = \frac{1}{\gamma} \inf_x \mathcal{R}_x(y).$$

Each conditional distribution $\mathcal{R}(x)$ can then be written as a mixture

$$\mathcal{R}(x) = \gamma Q^B + (1 - \gamma) \text{LO}(x),$$

where $\text{LO}(x)$ is the left-over distribution associated with input x . Operationally, sampling from $\mathcal{R}(x)$ is equivalent to drawing from Q^B with probability γ , and from $\text{LO}(x)$ otherwise.

For technical convenience, we also consider the *generalized blanket distribution* $\bar{Q}^B$ of $\mathcal{R}$ [SCW26b]:

$$\bar{Q}^B(y) = \begin{cases} \inf_x \mathcal{R}_x(y), & y \in \mathbb{Y}, \\ 1 - \gamma, & y = \perp, \end{cases}$$

where $\perp$ is a placeholder symbol representing “non-blanket” outcomes. The effective output space is thus extended to $\bar{\mathbb{Y}} = \mathbb{Y} \cup \{\perp\}$.

Theorem 15 (Blanket reduction [SCW26b]). *Let $\mathbf{Z}^r = \mathcal{S}(\{Y_1\} \cup \{Y'_i : i = 2, \dots, n\})$, where $Y'_i \stackrel{i.i.d.}{\sim} \bar{Q}^B$ and $\mathcal{S}$ is a uniform random shuffler. Then there exists a post-processing function f^B such that*

$$\forall \mathbf{x}_{-1} \in \mathbb{X}^{n-1} : f^B(\mathbf{Z}^r, \{\text{LO}(x_i)\}_{i=2}^n) \stackrel{d}{=} \mathbf{Z} \mid (\mathbf{X}_{-1} = \mathbf{x}_{-1}),$$

i.e., conditioned on $\mathbf{X}_{-1}$, the distribution of $\mathbf{Z}$ can be generated from $\mathbf{Z}^r$ via post-processing.

Corollary 3. *By the data-processing inequality, post-processing cannot increase mutual information. Consequently,*

$$\forall \mathbf{x}_{-1} \in \mathbb{X}^{n-1} : I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1} = \mathbf{x}_{-1}) \leq I(X_1; \mathbf{Z}^r \mid \mathbf{X}_{-1} = \mathbf{x}_{-1}).$$

6.2.2 Main Lemma in Canonical Shuffle DP Model

We now analyze $I(X_1; \mathbf{Z}^r \mid \mathbf{X}_{-1} = \mathbf{x}_{-1})$ in the canonical setting of Shuffle DP model. When conditioning on $\mathbf{X}_{-1} = \mathbf{x}_{-1}$, it is sufficient to work with $X_1 \sim P$ where P is the conditional probability distribution on $\mathbf{X}_{-1} = \mathbf{x}_{-1}$.

It is well known that producing a histogram of the outputs is equivalent (up to post-processing) to applying a random permutation to the multiset of messages; each representation can be obtained from the other [Che21]. For clarity, we work with the histogram view in the proof of Lemma 1.

Proof of Lemma 1. For an integer $s \geq 1$, let $M^{(s)} \sim \text{Mult}(s; Q)$ denote the *histogram* of s independent Q -samples. Thus $M^{(s)} = (M_y^{(s)})_{y \in \mathbb{Y}} \in \mathbb{N}^{|\mathbb{Y}|}$ satisfies $\sum_y M_y^{(s)} = s$. Assume $M^{(s)}$ is independent of the “signal” (X_1, Y_1) , where $X_1 \sim P$ and $Y_1 \mid X_1 = x \sim \mathcal{R}_x$. Define the *mixed* $(s+1)$ -sample histogram

$$C^{(s)} := e_{Y_1} + M^{(s)},$$

where $e_{Y_1} \in \mathbb{N}^{|\mathbb{Y}|}$ is the one-hot vector indicating the single draw Y_1 .

Given $C^{(s)} = e_{Y_1} + M^{(s)}$, Bayes’ rule implies (Fact 1 in Appendix F)

$$\Pr(X_1 = x \mid C^{(s)}) \propto P(x) T_x^{(s)}, \quad T_x^{(s)} := \sum_y C_y^{(s)} w_x(y).$$

We write

$$T_x^{(s)} = (s+1) + A_x^{(s)}, \quad A_x^{(s)} := \underbrace{\sum_y (M_y^{(s)} - sQ(y)) w_x(y)}_{\text{multinomial fluctuation}} + \underbrace{(w_x(Y_1) - 1)}_{\text{single-draw term}}.$$

Define

$$\varepsilon_x^{(s)} := \frac{A_x^{(s)}}{s+1}, \quad \bar{\varepsilon}^{(s)} := \sum_u P(u) \varepsilon_u^{(s)},$$

so that

$$\pi(x \mid C^{(s)}) = \frac{P(x) (1 + \varepsilon_x^{(s)})}{1 + \bar{\varepsilon}^{(s)}}.$$

A discrete Taylor expansion of $D(\pi \| P)$ around P yields (Lemma 5 in Appendix F)

$$D(\pi(\cdot \mid C^{(s)}) \| P) = \frac{1}{2} \sum_x P(x) (\varepsilon_x^{(s)} - \bar{\varepsilon}^{(s)})^2 + R(C^{(s)}), \quad |R(C^{(s)})| \leq K \|\varepsilon^{(s)}\|_3^3, \quad (22)$$

where $\|\varepsilon^{(s)}\|_3^3 := \sum_x P(x) |\varepsilon_x^{(s)}|^3$ and K is a constant. Taking expectations and using $I(X_1; C^{(s)}) = \mathbb{E}[D(\pi \| P)]$, we obtain

$$I(X_1; C^{(s)}) = \frac{1}{2} \mathbb{E} \left[\sum_x P(x) (\varepsilon_x^{(s)} - \bar{\varepsilon}^{(s)})^2 \right] + \mathbb{E} R(C^{(s)}). \quad (23)$$

Since $\bar{\varepsilon}^{(s)} = \sum_x P(x) \varepsilon_x^{(s)}$,⁷

$$\sum_x P(x) (\varepsilon_x^{(s)} - \bar{\varepsilon}^{(s)})^2 = \frac{1}{(s+1)^2} \text{Var}_{x \sim P}(A_x^{(s)}),$$

Hence

$$I(X_1; C^{(s)}) = \frac{1}{2(s+1)^2} \mathbb{E} [\text{Var}_{x \sim P}(A_x^{(s)})] + \mathbb{E} R(C^{(s)}). \quad (24)$$

⁷It is important to clarify the meaning of the notation $\text{Var}_{x \sim P}(\cdot)$ appearing in this expression. Here the randomness is *not* in the input variable X_1 . Instead, once the histogram $C^{(s)}$ is fixed, each quantity $A_x^{(s)}$ becomes a deterministic function of x , and the distribution $P(x)$ is used only as a weighting measure on the index set $\mathbb{X}$. Thus

$$\text{Var}_{x \sim P}(A_x^{(s)}) := \sum_x P(x) \left(A_x^{(s)} - \sum_u P(u) A_u^{(s)} \right)^2$$

is simply the variance of the *function* $x \mapsto A_x^{(s)}$ when the index x is sampled from P . With this interpretation, the identity above is just an algebraic rewriting of $\bar{\varepsilon}^{(s)} = \sum_x P(x) \varepsilon_x^{(s)}$, together with $\varepsilon_x^{(s)} = A_x^{(s)} / (s+1)$.

According to Lemma 6 in Appendix F,

$$\mathbb{E}[\text{Var}_{x \sim P}(A_x^{(s)})] = s(\bar{\chi}^2 - \chi^2(\bar{P} \| Q)) + O(1). \quad (25)$$

From (22) and (23), it remains to bound $\mathbb{E}\|\varepsilon_x^{(s)}\|_3^3$. Since $\varepsilon_x^{(s)} = A_x^{(s)}/(s+1)$ and $A_x^{(s)}$ is the sum of s centered bounded multinomial increments plus the bounded term $w_x(Y_1) - 1$, a standard moment bound (e.g., Rosenthal inequality) implies $\mathbb{E}|A_x^{(s)}|^3 = O(s^{3/2})$ uniformly in x . Hence

$$\mathbb{E}\|\varepsilon_x^{(s)}\|_3^3 = \sum_x P(x) \mathbb{E} \left[\left| \frac{A_x^{(s)}}{s+1} \right|^3 \right] = O((s+1)^{-3/2}),$$

and therefore $\mathbb{E} R(C^{(s)}) = O((s+1)^{-3/2})$.

Substituting (25) into (24) yields

$$I(X_1; C^{(s)}) = \frac{1}{2(s+1)^2} \left\{ s(\bar{\chi}^2 - \chi^2(\bar{P} \| Q)) + O(1) \right\} + O((s+1)^{-3/2}) = \frac{\bar{\chi}^2 - \chi^2(\bar{P} \| Q)}{2(s+1)} + O((s+1)^{-3/2}),$$

which is precisely (1). $\square$

6.2.3 Optimal Blanket Noise for Minimizing Input Leakage

Theorem 16. *In the canonical shuffle DP model, the noise distribution Q that asymptotically minimizes the input leakage $I(X_1; \mathbf{Z})$ is given by:*

$$Q(y) \propto \sqrt{\sum_{x \in \mathbb{X}} P(x) (\mathcal{R}_x(y) - \bar{P}(y))^2}.$$

Proof. By the definition of chi-square divergence, we have:

$$\begin{aligned} \sum_{x \in \mathbb{X}} P(x) \chi^2(\mathcal{R}_x \| Q) - \chi^2(\bar{P} \| Q) &= \sum_x P(x) \left[\sum_y \frac{\mathcal{R}_x(y)^2}{Q(y)} - 1 \right] - \left[\sum_y \frac{\bar{P}(y)^2}{Q(y)} - 1 \right] \\ &= \sum_y \frac{1}{Q(y)} \left[\sum_x P(x) \mathcal{R}_x(y)^2 - \bar{P}(y)^2 \right] \\ &= \sum_y \frac{\sum_x P(x) (\mathcal{R}_x(y) - \bar{P}(y))^2}{Q(y)}. \end{aligned}$$

Note that the numerator $\sum_x P(x) (\mathcal{R}_x(y) - \bar{P}(y))^2$ is the variance of the randomized response at output y with respect to the input distribution P . To minimize this expression subject to $\sum_y Q(y) = 1$, we invoke the Cauchy-Schwarz inequality:

$$\left(\sum_y \frac{a_y}{Q(y)} \right) \left(\sum_y Q(y) \right) \geq \left(\sum_y \sqrt{a_y} \right)^2,$$

where $a_y := \sum_x P(x) (\mathcal{R}_x(y) - \bar{P}(y))^2$. The lower bound is attained if and only if $Q(y) \propto \sqrt{a_y}$, which yields the optimal distribution. $\square$

Interpretation. This result admits a highly intuitive interpretation. For instance, if there exists an output $y^* \in \mathbb{Y}$ such that $\mathcal{R}_x(y^*)$ is invariant across all inputs $x \in \mathbb{X}$, then observing $Y_1 = y^*$ is entirely uninformative regarding X_1 (i.e., the posterior distribution of X_1 remains identical to its prior). Consequently, there is no need to allocate any blanket noise probability at y^* . Theorem 16 formalizes this intuition: the optimal noise distribution Q should prioritize outputs where the variance of $\mathcal{R}_x(y)$ with respect to $x \sim P$ is higher. Specifically, the optimal $Q(y)$ is directly proportional to the standard deviation of these local probabilities.

6.2.4 Analysis of Canonical Shuffle DP with Generalized Blanket

Proof of Lemma 2. By the definition of the Pearson χ^2 divergence, we have:

$$\begin{aligned} \mathbb{E}_{x \sim P}[\chi^2(\mathcal{R}_x \| \bar{Q}^B)] - \chi^2(\bar{P} \| \bar{Q}^B) &= \sum_{y \in \mathbb{Y}} \frac{\mathbb{E}_{x \sim P}[\mathcal{R}_x(y)^2]}{\bar{Q}^B(y)} - \sum_{y \in \mathbb{Y}} \frac{\bar{P}(y)^2}{\bar{Q}^B(y)} \\ &= \sum_{y \in \mathbb{Y}} \frac{\mathbb{E}_{x \sim P}[\mathcal{R}_x(y)^2] - \bar{P}(y)^2}{\bar{Q}^B(y)}. \end{aligned}$$

For fixed distributions $\bar{P}$ and $\bar{Q}^B$, we aim to maximize the above expression over all valid ε_0 -DP randomizers. The ε_0 -DP property implies the pointwise constraint $\bar{Q}^B(y) \leq \mathcal{R}_x(y) \leq e^{\varepsilon_0} \bar{Q}^B(y)$ for all $y \in \mathbb{Y}$ and $x \in \mathbb{X}$. Due to the convexity of the quadratic function $f(t) = t^2$, for fixed y , $\bar{P}(y)$, and $\bar{Q}^B(y)$, the term $\mathbb{E}_{x \sim P}[\mathcal{R}_x(y)^2] - \bar{P}(y)^2$ (which represents the variance of $\mathcal{R}_x(y)$ under P) is maximized when $\mathcal{R}_x(y)$ takes values at the boundaries of its feasible interval, i.e., $\mathcal{R}_x(y) \in \{\bar{Q}^B(y), e^{\varepsilon_0} \bar{Q}^B(y)\}$. In this extremal case, the distribution of $\mathcal{R}_x(y)$ is determined by the mean $\bar{P}(y)$, yielding:

$$\begin{aligned} \mathbb{E}_{x \sim P}[\mathcal{R}_x(y)^2] - \bar{P}(y)^2 &= \frac{e^{\varepsilon_0} \bar{Q}^B(y) - \bar{P}(y)}{(e^{\varepsilon_0} - 1) \bar{Q}^B(y)} \cdot \bar{Q}^B(y)^2 + \frac{\bar{P}(y) - \bar{Q}^B(y)}{(e^{\varepsilon_0} - 1) \bar{Q}^B(y)} \cdot e^{2\varepsilon_0} \bar{Q}^B(y)^2 - \bar{P}(y)^2 \\ &= \frac{\bar{Q}^B(y)}{e^{\varepsilon_0} - 1} \{e^{\varepsilon_0} \bar{Q}^B(y) - \bar{P}(y) + e^{2\varepsilon_0} [\bar{P}(y) - \bar{Q}^B(y)]\} - \bar{P}(y)^2. \end{aligned}$$

Substituting this back into the total sum, we obtain:

$$\sum_{y \in \mathbb{Y}} \frac{\mathbb{E}_{x \sim P}[\mathcal{R}_x(y)^2] - \bar{P}(y)^2}{\bar{Q}^B(y)} \leq \sum_{y \in \mathbb{Y}} \left[\frac{e^{\varepsilon_0} \bar{Q}^B(y) - \bar{P}(y) + e^{2\varepsilon_0} (\bar{P}(y) - \bar{Q}^B(y))}{e^{\varepsilon_0} - 1} - \frac{\bar{P}(y)^2}{\bar{Q}^B(y)} \right].$$

Let $\gamma = \sum_{y \in \mathbb{Y}} \bar{Q}^B(y)$. Since $\sum \bar{P}(y) = 1$, the expression simplifies to:

$$\frac{e^{\varepsilon_0} \gamma - 1 + e^{2\varepsilon_0} (1 - \gamma)}{e^{\varepsilon_0} - 1} - \sum_{y \in \mathbb{Y}} \frac{\bar{P}(y)^2}{\bar{Q}^B(y)} = e^{\varepsilon_0} + 1 - e^{\varepsilon_0} \gamma - \sum_{y \in \mathbb{Y}} \frac{\bar{P}(y)^2}{\bar{Q}^B(y)}.$$

By the Cauchy-Schwarz inequality, $\sum_{y \in \mathbb{Y}} \frac{\bar{P}(y)^2}{\bar{Q}^B(y)} \geq \frac{(\sum \bar{P}(y))^2}{\sum \bar{Q}^B(y)} = \frac{1}{\gamma}$, where the equality holds when $\bar{P}(y) = \bar{Q}^B(y)/\gamma$. Thus:

$$e^{\varepsilon_0} + 1 - e^{\varepsilon_0} \gamma - \sum_{y \in \mathbb{Y}} \frac{\bar{P}(y)^2}{\bar{Q}^B(y)} \leq e^{\varepsilon_0} + 1 - e^{\varepsilon_0} \gamma - \frac{1}{\gamma}.$$

To find the global maximum, we optimize with respect to γ . The function $g(\gamma) = e^{\varepsilon_0} + 1 - e^{\varepsilon_0} \gamma - \frac{1}{\gamma}$ reaches its maximum when $g'(\gamma) = -e^{\varepsilon_0} + \frac{1}{\gamma^2} = 0$, which gives $\gamma = e^{-\varepsilon_0/2}$. Substituting this value back, we have:

$$e^{\varepsilon_0} + 1 - e^{\varepsilon_0} \cdot e^{-\varepsilon_0/2} - e^{\varepsilon_0/2} = e^{\varepsilon_0} + 1 - 2e^{\varepsilon_0/2} = (e^{\varepsilon_0/2} - 1)^2.$$

This confirms the bound, and the tightness is established by the conditions for equality in the Cauchy-Schwarz and extremal value steps. $\square$

Theorem 17. *In the shuffle-DP setting, for any fixed $\mathbf{x}_{-1} \in \mathbb{X}^{n-1}$,*

$$I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1} = \mathbf{x}_{-1}) \leq \frac{(e^{\varepsilon_0/2} - 1)^2}{2n} + O(n^{-3/2}).$$

Taking expectation over $\mathbf{X}_{-1}$ yields

$$I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1}) \leq \frac{(e^{\varepsilon_0/2} - 1)^2}{2n} + O(n^{-3/2}).$$

Proof. Applying Lemma 1 with $s = n - 1$, Lemma 2 and then using Corollary 3 gives

$$I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1} = \mathbf{x}_{-1}) \leq I(X_1; \mathbf{Z}^r \mid \mathbf{X}_{-1} = \mathbf{x}_{-1}) \leq \frac{(e^{\varepsilon_0/2} - 1)^2}{2n} + O(n^{-3/2}). \quad \square$$

Corollary 4. *In the shuffle-DP setting, if X_1 is independent of $\mathbf{X}_{-1}$, then*

$$I(X_1; \mathbf{Z}) \leq \frac{(e^{\varepsilon_0/2} - 1)^2}{2n} + O(n^{-3/2}).$$

Proof. The argument follows the same reasoning as in Corollary 2. $\square$

Example 6.1. *Let $[k] := \{1, 2, \dots, k\}$ and denote by $\mathcal{U}_{[k]}$ the uniform distribution on $[k]$. For any $k \in \mathbb{N}$ and $\varepsilon_0 > 0$, the k -ary randomized response mechanism $k\text{RR}_{\varepsilon_0} : [k] \rightarrow [k]$ is defined as*

$$k\text{RR}_{\varepsilon_0}(x) = \begin{cases} x, & \text{with probability } \frac{e^{\varepsilon_0} - 1}{e^{\varepsilon_0} + k - 1}, \\ y \sim \mathcal{U}_{[k]}, & \text{with probability } \frac{k}{e^{\varepsilon_0} + k - 1}. \end{cases}$$

The generalized blanket distribution of $k\text{RR}_{\varepsilon_0}$ is

$$\bar{Q}^B = \frac{k}{e^{\varepsilon_0} + k - 1} \mathcal{U}_{[k]} + \frac{e^{\varepsilon_0} - 1}{e^{\varepsilon_0} + k - 1} \mathbb{1}_{\perp}.$$

For any input distribution P ,

$$\overline{\chi^2}(P \parallel \bar{Q}^B) := \mathbb{E}_{x \sim P}[\chi^2(\mathcal{R}_x \parallel \bar{Q}^B)] = \frac{e^{\varepsilon_0}}{e^{\varepsilon_0} + k - 1} (e^{\varepsilon_0} - 1).$$

By the Cauchy-Schwarz inequality, $\chi^2(\bar{P} \parallel \bar{Q}^B) \geq \frac{(\sum \bar{P}(y))^2}{\sum \bar{Q}^B(y)} - 1 = \frac{1}{\gamma} - 1 = \frac{e^{\varepsilon_0} - 1}{k}$. By Lemma 1, this implies the upper bound

$$\forall \mathbf{x}_{-1} : I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1} = \mathbf{x}_{-1}) \leq \frac{k - 1}{(e^{\varepsilon_0} + k - 1)k} \cdot \frac{(e^{\varepsilon_0} - 1)^2}{2n} + O(n^{-3/2}).$$

Now suppose that $X_2, X_3, \dots, X_n$ are i.i.d. uniform on $[k]$. Then the actual blanket distribution is $Q = k\text{RR}_{\varepsilon_0}(\mathcal{U}_{[k]}) = \mathcal{U}_{[k]}$. In this case, for any P ,

$$\overline{\chi^2}(P \parallel Q) := \mathbb{E}_{x \sim P}[\chi^2(\mathcal{R}_x \parallel Q)] = \frac{(k - 1)(e^{\varepsilon_0} - 1)^2}{(e^{\varepsilon_0} + k - 1)^2}.$$

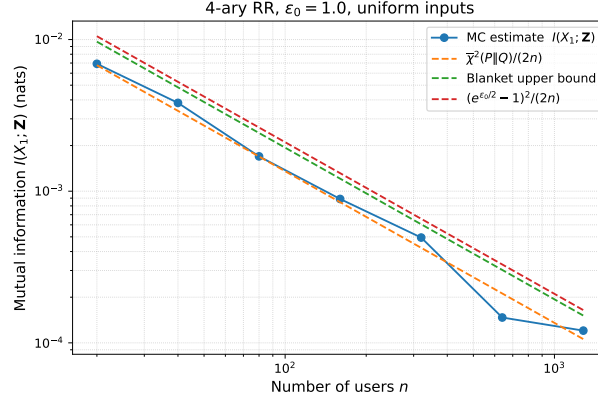


Figure 5: Mutual information in the shuffle-DP setting: numerical estimates vs. asymptotic bounds.

When $X_1 \sim P = \mathcal{U}_{[k]}$, the mixture $\bar{P}$ coincides with $Q = \mathcal{U}_{[k]}$, so $\chi^2(\bar{P}||Q) = 0$. Applying Lemma 1 again yields

$$I(X_1; \mathbf{Z}) = \frac{(k-1)(e^{\varepsilon_0} - 1)^2}{2(e^{\varepsilon_0} + k - 1)^2 n} + O(n^{-3/2}),$$

showing that there exist input distributions (X_i) for which this rate is asymptotically attained.

Finally, we run a numerical experiment in the shuffle-DP setting using the 4-ary randomized response mechanism with uniformly distributed inputs X_i ($i = 1, 2, \dots, n$). The empirical mutual information $I(X_1; \mathbf{Z})$, estimated via Monte Carlo with 10^5 samples, closely matches the asymptotic prediction $\frac{\chi^2(P||Q) - \chi^2(\bar{P}||Q)}{2n} = \frac{\chi^2(P||Q)}{2n}$, and lies below both the blanket-decomposition upper bound $\frac{k-1}{(e^{\varepsilon_0} + k - 1)k} \cdot \frac{(e^{\varepsilon_0} - 1)^2}{2n}$ and the unified bound $\frac{(e^{\varepsilon_0/2} - 1)^2}{2n}$. These results provide empirical support for our asymptotic analysis.

Remark 3. The leakage $I(X_1; \mathbf{Z})$ can be large when the X_i are not mutually independent. For instance, under full coherence $X_1 = X_2 = \dots = X_n$ (each with marginal distribution P), the shuffle-DP model with the randomized response mechanism produces $\mathbf{Z}$ as n i.i.d. randomized reports of the same underlying symbol. By the law of large numbers, a simple majority-vote decoder recovers X_1 with probability tending to one, so the leakage is almost complete:

$$I(X_1; \mathbf{Z}) \longrightarrow H(X_1) \quad \text{as } n \rightarrow \infty.$$

7 Discussion

We employ the blanket decomposition to reduce the intractable shuffle-DP channel to a more analyzable canonical setting. Another well-known decomposition is the *clone decomposition* [FMT22], which states that for any fixed x_1 , the output distribution of any other input x can be written as

$$\forall x \in \mathbb{X}, \quad \mathcal{R}(x) = e^{-\varepsilon_0} \mathcal{R}(x_1) + (1 - e^{-\varepsilon_0}) \text{LO}(x).$$

Under this representation, with probability $e^{-\varepsilon_0}$ an arbitrary user effectively produces a “clone” of the user holding x_1 .

However, the clone decomposition is *not* suitable for analyzing $I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1})$. Indeed, after conditioning on $\mathbf{X}_{-1}$, the decomposition reduces the shuffled multiset to

$$1 + N \text{ samples from } \mathcal{R}(X_1), \quad N \sim \text{Bin}(n-1, e^{-\varepsilon_0}),$$

which, with overwhelming probability, reveals almost complete information about X_1 (see Remark 3). In contrast, the blanket distribution is *independent* of X_1 , which preserves the anonymity needed for a meaningful information-theoretic analysis of $I(X_1; \mathbf{Z})$.

Nonetheless, the clone decomposition remains very useful for understanding $I(Y_1; \mathbf{Z} \mid \mathbf{X}_{-1})$. When our goal is to bound the leakage about the *message* Y_1 rather than the underlying input X_1 , the clone representation reduces the shuffle-DP mechanism to the *canonical shuffle-only model* (Section 3.1) with

$$P = Q \quad \text{and} \quad N = 1 + \text{Bin}(n-1, e^{-\varepsilon_0}).$$

Applying Theorem 10 to this reduced instance gives

$$I(Y_1; \mathbf{Z} \mid \mathbf{X}_{-1}) \leq \mathbb{E}_{N \sim \text{Bin}(n-1, e^{-\varepsilon_0})} \left[\frac{m-1}{2(N+1)} + O(N^{-3/2}) \right] = \frac{(m-1)e^{\varepsilon_0}}{2n} + O(n^{-3/2}),$$

where $m = |\mathbb{Y}|$. Thus the clone decomposition, although unsuitable for bounding the leakage about *inputs*, provides a clean characterization of message-level privacy leakage.

It is worth noting that the blanket decomposition also applies to the analysis of $I(Y_1; \mathbf{Z} \mid \mathbf{X}_{-1})$, in which case the corresponding reduced setting is the canonical shuffle-only model with $P \neq Q$. As shown in [SCW26b], conditioned on $\mathbf{X}_{-1}$, there exists a post-processing function from the mixture induced by the clone reduction to that induced by the blanket reduction. Consequently, the blanket decomposition yields a tighter upper bound, although the resulting expressions are algebraically more involved than those obtained from the clone decomposition.

8 Conclusion

This work establishes the first formal link between the canonical setting of the shuffle-only model and cryptographic primitives, such as the Ideal Cipher Model and Honey Encryption, which naturally motivates our investigation into optimal blanket noise design. Across four classical privacy metrics, we demonstrate that the optimal noise distribution Q generally deviates from the input distribution P ; notably, in several regimes, the optimal Q admits a remarkably concise and elegant form.

Within the shuffle-DP framework, we characterize the optimal design of Q that minimizes mutual information leakage and derive a tight privacy amplification bound for the single-message shuffle model. We anticipate that these findings will significantly deepen the theoretical understanding of shuffling as a privacy-enhancing mechanism. To this end, we pose the following open questions:

- **Alternative Analyses for the General Shuffle-Only Setting.** The general shuffle-only scenario remains both intriguing and technically challenging. In this work, we employ the *blanket decomposition* technique to reduce the analysis to a simpler, tractable form. This reduction relies on the existence of a common component among the remaining users' distributions to obscure the target user. An open question is whether alternative analytical frameworks can be developed to study this setting directly without relying on such a decomposition.
- **Tight and Non-Asymptotic Bounds on Information Leakage.** The residual terms in our asymptotic bounds depend on the distributions P, Q in the shuffle-only setting and on the local randomizer $\mathcal{R}$ in the shuffle-DP setting. This dependence appears inevitable in

the shuffle-only case, as it is inherently related to the support size of P and Q (see Theorem 10). However, for the shuffle-DP regime, it remains unclear whether one can establish a *non-asymptotic*, distribution-independent upper bound on $I(X_1; \mathbf{Z} \mid \mathbf{X}_{-1})$. Resolving this question would offer a deeper information-theoretic understanding of the shuffle model and its privacy amplification behavior.

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A Proof of Theorem 7

Proof. Recall that the shuffling process permutes the users’ messages according to a random permutation σ , uniformly sampled from the permutation group $\mathcal{S}_n$. Let $\mathbf{y} = (y_1, \dots, y_n)$ denote the original (unshuffled) message sequence. By definition, the observed shuffled output satisfies $z_i = y_{\sigma(i)}$ for each i .

The posterior probability of $K = k$ given $\mathbf{Z} = \mathbf{z}$ can be written as

$$\Pr[K = k \mid \mathbf{Z} = \mathbf{z}] = \Pr[K = k \mid \mathbf{y} = \mathbf{z}_{\sigma^{-1}}], \quad (26)$$

since knowing $\mathbf{Z} = \mathbf{z}$ is equivalent to knowing $\mathbf{y}$ under the inverse permutation.

Using the definition of conditional probability,

$$\Pr[K = k \mid \mathbf{y} = \mathbf{z}_{\sigma^{-1}}] = \frac{\Pr[K = k \wedge \mathbf{y} = \mathbf{z}_{\sigma^{-1}}]}{\Pr[\mathbf{y} = \mathbf{z}_{\sigma^{-1}}]}. \quad (27)$$

We next express the numerator and denominator using the law of total probability over all possible permutations $\sigma_1 \in \mathcal{S}_n$:

$$\Pr[K = k \mid \mathbf{Z} = \mathbf{z}] = \frac{\Pr[K = k] \cdot \Pr[\mathbf{y} = \mathbf{z}_{\sigma^{-1}} \mid K = k]}{\sum_{\sigma_1 \in \mathcal{S}_n} \Pr[\sigma = \sigma_1] \cdot \Pr[\mathbf{y} = \mathbf{z}_{\sigma^{-1}} \mid \sigma = \sigma_1]}. \quad (28)$$

Two key facts are used here:

1. The position K is uniformly distributed, i.e., $\Pr[K = k] = 1/n$.
2. Conditioned on $K = k$, the element at position z_k is drawn from P , whereas the remaining z_j ’s are drawn from Q .

Hence, the joint likelihood becomes

$$\Pr[\mathbf{y} = \mathbf{z}_{\sigma^{-1}} \mid K = k] = P(z_k) \prod_{j \neq k} Q(z_j) = \frac{P(z_k)}{Q(z_k)} \prod_{j=1}^n Q(z_j). \quad (29)$$

Plugging (29) into (28), we obtain

$$\Pr[K = k \mid \mathbf{Z} = \mathbf{z}] = \frac{\frac{1}{n} \frac{P(z_k)}{Q(z_k)} \prod_{j=1}^n Q(z_j)}{\sum_{i=1}^n \frac{1}{n} \frac{P(z_i)}{Q(z_i)} \prod_{j=1}^n Q(z_j)}. \quad (30)$$

After canceling the common factors $\frac{1}{n}$ and $\prod_{j=1}^n Q(z_j)$ in the numerator and denominator, we obtain the normalized posterior:

$$\Pr[K = k \mid \mathbf{Z} = \mathbf{z}] = \frac{\frac{P(z_k)}{Q(z_k)}}{\sum_{i=1}^n \frac{P(z_i)}{Q(z_i)}}. \square$$

B Details in the Proof of Theorem 10

This section clarifies several technical steps used in the proof. Let $f(t) := t \log t$ and fix an index i . Define

$$g(t) = p_i \log p_i + (\log p_i + 1)(t - p_i) + \frac{1}{2p_i}(t - p_i)^2.$$

By the second-order Taylor expansion of f around $t = p_i$ with Lagrange remainder, we have

$$f(t) = g(t) + R(t), \tag{31}$$

where

$$R(t) = \frac{f^{(3)}(\theta)}{6}(t - p_i)^3 = -\frac{1}{6\theta^2}(t - p_i)^3, \quad \theta \in (t, p_i).$$

As $t \rightarrow 0$, no uniform bound on $1/\theta^2$ is available. To handle this rigorously, define the event

$$\mathcal{E} := \left\{ \frac{X}{n} \leq \frac{p_i}{2} \right\}.$$

By the Chernoff bound, $\Pr[\mathcal{E}] = \text{negl}(n)$,⁸ i.e., the probability of $\mathcal{E}$ decays faster than any inverse polynomial in n .

For $t \geq p_i/2$, the remainder is uniformly bounded as

$$|R(t)| \leq \frac{2}{3p_i^2} |t - p_i|^3.$$

Therefore,

$$\begin{aligned} \mathbb{E}_{X \sim \text{Bin}(n, p_i)} \left[f\left(\frac{X}{n}\right) \right] &= \mathbb{E} \left[g\left(\frac{X}{n}\right) \right] + \mathbb{E} \left[R\left(\frac{X}{n}\right) \right] \\ &= p_i \log p_i + \frac{1 - p_i}{2n} + \sum_{x > np_i/2} p_X(x) R\left(\frac{x}{n}\right) + \sum_{x \leq np_i/2} p_X(x) R\left(\frac{x}{n}\right) \\ &= p_i \log p_i + \frac{1 - p_i}{2n} + O\left(\sum_{x > np_i/2} p_X(x) \left| R\left(\frac{x}{n}\right) \right| \right) + O\left(\sum_{x \leq np_i/2} p_X(x) \left| R\left(\frac{x}{n}\right) \right| \right) \\ &= p_i \log p_i + \frac{1 - p_i}{2n} + O\left(\mathbb{E} \left[\left| \frac{X}{n} - p_i \right|^3 \right] \right) + \text{negl}(n) \cdot M. \end{aligned}$$

where $M = \max_{t \in [0, 1]} |R(t)| = \max_{t \in [0, 1]} |f(t) - g(t)| = O(1)$.

To bound $O(\mathbb{E}[\left|\frac{X}{n} - p_i\right|^3])$, we use the following inequality.

Theorem 18 (Rosenthal inequality [CS20]). *Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}|X_i|^p < \infty$ for some $p \geq 2$, and set $Y_i := X_i - \mathbb{E}X_i$. Then there exists a constant $C_p > 0$ depending only on p such that*

$$\mathbb{E} \left| \sum_{i=1}^n Y_i \right|^p \leq C_p \max \left\{ \left(\sum_{i=1}^n \mathbb{E}|Y_i|^2 \right)^{p/2}, \sum_{i=1}^n \mathbb{E}|Y_i|^p \right\}.$$

Consequently, there is $C'_p > 0$ depending only on p such that

$$\mathbb{E} \left| \sum_{i=1}^n X_i \right|^p \leq C'_p \left[\left(\sum_{i=1}^n \text{Var}[X_i] \right)^{p/2} + \sum_{i=1}^n \mathbb{E}|X_i - \mathbb{E}X_i|^p + \sum_{i=1}^n \mathbb{E}|X_i|^p \right].$$

⁸We use the standard notation $\phi(n) = \text{negl}(n)$ to mean that for every $t \in \mathbb{N}$, $\lim_{n \rightarrow \infty} n^t \phi(n) = 0$.

Let $A_1, \dots, A_n$ be i.i.d. Bernoulli random variables with mean p_i , and write $X = \sum_{j=1}^n A_j$. Then

$$\left| \frac{X}{n} - p_i \right| = \frac{1}{n} \left| \sum_{j=1}^n (A_j - p_i) \right|.$$

Applying Theorem 18 with $p = 3$,

$$\begin{aligned} \mathbb{E} \left| \sum_{j=1}^n (A_j - p_i) \right|^3 &= O \left(\left(\sum_{j=1}^n \mathbb{E}[(A_j - p_i)^2] \right)^{3/2} + \sum_{j=1}^n \mathbb{E}|A_j - p_i|^3 \right) \\ &= O \left((n p_i (1 - p_i))^{3/2} + n [p_i (1 - p_i)^3 + (1 - p_i) p_i^3] \right) = O(n^{3/2}). \end{aligned}$$

Hence

$$\mathbb{E} \left| \frac{X}{n} - p_i \right|^3 = \frac{1}{n^3} \mathbb{E} \left| \sum_{j=1}^n (A_j - p_i) \right|^3 = O(n^{-3/2}).$$

Combining the bounds above yields the desired estimate.

C Details in the Proof of Theorem 1

Lemma 4. *Let (S, T) be random variables (depending on the parameter n) with $S > 0$ a.s., and let*

$$\mu_S := \mathbb{E}[S], \quad \mu_T := \mathbb{E}[T].$$

Assume there exist constants $c_1, c_2, c_3, c_4, c_5 > 0$, independent of n , such that

$$c_1 \leq S \leq c_2 n, \quad \Pr[S \leq c_3 n] = \text{negl}(n), \quad |T| \leq c_4 n, \quad \mathbb{E}(|S - \mu_S|^3 + |T - \mu_T|^3) \leq c_5 n^{3/2}.$$

Let $f(s, t) := t/s$. Then

$$\mathbb{E} \left[\frac{T}{S} \right] = \frac{\mu_T}{\mu_S} - \frac{\text{Cov}(T, S)}{\mu_S^2} + \frac{\mu_T \text{Var}(S)}{\mu_S^3} + O(n^{-3/2}). \quad (32)$$

Proof. Set

$$\Delta_S := S - \mu_S, \quad \Delta_T := T - \mu_T,$$

so that $\mathbb{E}[\Delta_S] = \mathbb{E}[\Delta_T] = 0$, $\text{Var}(S) = \mathbb{E}[\Delta_S^2]$, $\text{Var}(T) = \mathbb{E}[\Delta_T^2]$, and $\text{Cov}(S, T) = \mathbb{E}[\Delta_S \Delta_T]$.

For $f(s, t) = t/s$ on $(0, \infty) \times \mathbb{R}$,

$$f_s(s, t) = -\frac{t}{s^2}, \quad f_t(s, t) = \frac{1}{s}, \quad f_{ss}(s, t) = \frac{2t}{s^3}, \quad f_{st}(s, t) = -\frac{1}{s^2}, \quad f_{tt}(s, t) = 0.$$

Consider the quadratic Taylor polynomial at (μ_S, μ_T) :

$$\begin{aligned} g(S, T) &= f(\mu_S, \mu_T) + f_s(\mu_S, \mu_T) \Delta_S + f_t(\mu_S, \mu_T) \Delta_T + \\ &\quad \frac{1}{2} (f_{ss}(\mu_S, \mu_T) \Delta_S^2 + 2f_{st}(\mu_S, \mu_T) \Delta_S \Delta_T + f_{tt}(\mu_S, \mu_T) \Delta_T^2). \end{aligned}$$

By the two-variable Taylor formula with Lagrange remainder,

$$f(S, T) = g(S, T) + R, \quad (33)$$

where the third-order remainder satisfies

$$|R| \leq \frac{1}{6} \left(\sup_{\mathcal{N}} \|D^3 f\| \right) (|\Delta_S| + |\Delta_T|)^3, \quad (34)$$

with the supremum taken over the line segment $\mathcal{N} = \{(\mu_S, \mu_T) + \theta(\Delta_S, \Delta_T) : \theta \in [0, 1]\}$, and $\|D^3 f\|$ any operator norm of the array of third partials. The nonzero third partials are $f_{sss}(s, t) = -6t/s^4$ and $f_{sst}(s, t) = 2/s^3$.

From (33) and centering, the linear terms vanish, giving

$$\begin{aligned} \mathbb{E}\left[\frac{T}{S}\right] &= f(\mu_S, \mu_T) + \frac{1}{2}(f_{ss}(\mu_S, \mu_T) \text{Var}(S) + 2f_{st}(\mu_S, \mu_T) \text{Cov}(S, T) + f_{tt}(\mu_S, \mu_T) \text{Var}(T)) + \mathbb{E}[R] \\ &= \frac{\mu_T}{\mu_S} - \frac{\text{Cov}(T, S)}{\mu_S^2} + \frac{\mu_T \text{Var}(S)}{\mu_S^3} + \mathbb{E}[R]. \end{aligned}$$

Bounding the remainder $\mathbb{E}[R]$. Let $\mathcal{E} = \{S \geq c_3 n\}$. On $\mathcal{E}$, since $|T| \leq c_4 n$ and $S \geq c_3 n$, the nonzero third partial derivatives of $f(s, t) = t/s$ satisfy

$$\sup_{\mathcal{N}} \|D^3 f\| = \max \left\{ \sup_{\mathcal{N}} |f_{sss}|, \sup_{\mathcal{N}} |f_{sst}| \right\} \leq \max \left\{ \frac{6c_4}{c_3^4 n^3}, \frac{2}{c_3^3 n^3} \right\} = O(n^{-3}).$$

Hence, by (34),

$$\mathbb{E}[|R| \mathbf{1}_{\mathcal{E}}] \leq O(n^{-3}) \mathbb{E}[(|\Delta_S| + |\Delta_T|)^3] = O(n^{-3}) \cdot O(n^{3/2}) = O(n^{-3/2}),$$

using the moment assumption.

On $\mathcal{E}^c$, we use a crude polynomial bound. Since $S \geq c_1 > 0$ and $|T| \leq c_4 n$, we have

$$\begin{aligned} |f(S, T)| &\leq \frac{c_4}{c_1} n, \\ |g(S, T)| &\leq |f(\mu_S, \mu_T)| + |f_s(\mu_S, \mu_T) \Delta_S| + |f_t(\mu_S, \mu_T) \Delta_T| \\ &\quad + \frac{1}{2} (|f_{ss}(\mu_S, \mu_T) \Delta_S^2| + 2|f_{st}(\mu_S, \mu_T) \Delta_S \Delta_T| + |f_{tt}(\mu_S, \mu_T) \Delta_T^2|) \\ &\leq \frac{c_4}{c_1} n + \frac{c_4 n}{c_1^2} \cdot c_2 n + \frac{c_4 n}{c_1} + \frac{1}{2} \left(\frac{2c_4 n}{c_1^3} \cdot (c_2 n)^2 + \frac{2}{c_1^2} \cdot c_2 c_4 n^2 \right) \\ &= O(n^3). \end{aligned}$$

Consequently,

$$\mathbb{E}[|R| \mathbf{1}_{\mathcal{E}^c}] \leq \mathbb{E}[(|f(S, T)| + |g(S, T)|) \mathbf{1}_{\mathcal{E}^c}] \leq O(n^3) \Pr[\mathcal{E}^c] = \text{negl}(n).$$

Combining the two parts yields $\mathbb{E}[R] = O(n^{-3/2})$, which establishes (32). $\square$

It remains to verify that the assumptions of Lemma 4 hold for the random variables

$$S := \sum_{j=1}^n w(Z_j), \quad T := \sum_{j=1}^n w(Z_j) \log w(Z_j), \quad \mathbf{Z} = (Z_1, \dots, Z_n).$$

Among the n summands in S , exactly one term is $W^{(P)} := w(Y_1)$ with $Y_1 \sim P$, while the remaining $n - 1$ terms are i.i.d. $W_i^{(Q)} := w(Y_i)$ with $Y_i \sim Q$.

Let

$$w_{\min}^{(P)} := \min_{y \in \text{Supp}(P)} \frac{P(y)}{Q(y)} > 0, \quad w_{\max}^{(P)} := \max_{y \in \text{Supp}(P)} \frac{P(y)}{Q(y)},$$

$$w_{\max}^{(Q)} := \max_{y \in \text{Supp}(Q)} \frac{P(y)}{Q(y)}, \quad M := \max_y |w(y) \log w(y)|.$$

Then, deterministically,

$$S \geq w_{\min}^{(P)}, \quad S \leq w_{\max}^{(P)} + (n-1) w_{\max}^{(Q)} \leq n \cdot \max\{w_{\max}^{(P)}, w_{\max}^{(Q)}\}, \quad |T| \leq nM.$$

Moreover,

$$\mu_S := \mathbb{E}[S] = \mathbb{E}_P[W] + (n-1)\mathbb{E}_Q[W] = (1 + \chi^2) + (n-1) \cdot 1 = n + \chi^2,$$

so $\mu_S = \Theta(n)$. Since $\{w(Z_j)\}_{j=1}^n$ are bounded and independent, a standard Chernoff bound yields, for some constant $c > 0$,

$$\Pr[S \leq \tfrac{1}{2}\mu_S] \leq e^{-cn} = \text{negl}(n).$$

For the third centered moments, by the Rosenthal inequality (with $p = 3$),

$$\mathbb{E}(|S - \mu_S|^3) = O(n^{3/2}), \quad \mathbb{E}(|T - \mu_T|^3) = O(n^{3/2}),$$

since $\text{Var } S = \Theta(n)$ and $\text{Var } T = \Theta(n)$. These bounds verify the hypotheses of Lemma 4 with constants independent of n , completing the check.

D Canonical Shuffle-Only Setting: $P \not\ll Q$

We now extend the analysis from the case $P \ll Q$ to the more general situation where P may assign positive mass to events outside the support of Q . Let $\mathbb{Y} = \mathbb{Y}_1 \cup \mathbb{Y}_2$, where

$$\mathbb{Y}_2 = \{y \mid P(y) > 0 \text{ and } Q(y) = 0\}, \quad \mathbb{Y}_1 = \mathbb{Y} \setminus \mathbb{Y}_2.$$

Define the indicator random variable $T = \mathbb{1}[Y_1 \in \mathbb{Y}_2]$. When $T = 1$, the event $Y_1 \in \mathbb{Y}_2$ implies that Y_1 takes a value impossible under Q , which makes the shuffled output $\mathbf{Z}$ reveal Y_1 completely. Conditioning on $T = 0$, the distribution reduces to a restricted case $P' = P|_{\mathbb{Y}_1}$ such that $P' \ll Q$. Let $\beta = \Pr[T = 1] = P(\mathbb{Y}_2)$, so that $P = (1 - \beta)P|_{\mathbb{Y}_1} + \beta P|_{\mathbb{Y}_2}$.

The following results generalize the asymptotic expansions obtained in the absolutely continuous case.

Theorem 19. *In the canonical shuffle-only model, when $P \not\ll Q$, the mutual information between the shuffle position K and the shuffled output $\mathbf{Z}$ satisfies*

$$I(K; \mathbf{Z}) = \beta \log n + (1 - \beta) D_{\text{KL}}(P' \| Q) - (1 - \beta) \frac{\chi^2(P' \| Q)}{2n} + O(n^{-3/2}), \quad n \rightarrow \infty,$$

where $\beta = P(\mathbb{Y}_2)$, $P' = P|_{\mathbb{Y} \setminus \mathbb{Y}_2}$, and $\mathbb{Y}_2 = \{y \mid P(y) > 0 \text{ and } Q(y) = 0\}$.

Proof. By the chain rule, $I(K; \mathbf{Z}, T) = I(K; T) + I(K; \mathbf{Z} | T) = I(K; \mathbf{Z}) + I(K; T | \mathbf{Z})$, we obtain

$$I(K; \mathbf{Z}) = I(K; \mathbf{Z} | T) + I(K; T) - I(K; T | \mathbf{Z}).$$

Observe that T is deterministically recoverable from $\mathbf{Z}$ (one simply checks whether $\mathbf{Z}$ contains any element from $\mathbb{Y}_2$). Hence $I(K; T | \mathbf{Z}) = 0$. Furthermore, K and T are independent: T is an intermediate random variable generated in the process $Y_1 \sim P$, whereas K is drawn independently by the shuffler. Thus $I(K; T) = 0$.

Therefore,

$$I(K; \mathbf{Z}) = I(K; \mathbf{Z} | T) = I(K; \mathbf{Z} | T = 0) \Pr[T = 0] + I(K; \mathbf{Z} | T = 1) \Pr[T = 1].$$

When $T = 1$, i.e., $Y_1 \in \mathbb{Y}_2$, the shuffle output deterministically reveals the position of Y_1 in $\mathbf{Z}$. Hence $I(K; \mathbf{Z} | T = 1) = \log n$.

When $T = 0$, the random variable Y_1 follows the restricted distribution $P' \ll Q$. In this case, all previously derived expansions for $I(K; \mathbf{Z})$ under $P \ll Q$ (see Theorem 1) directly apply to P' :

$$I(K; \mathbf{Z} | T = 0) = D_{\text{KL}}(P' \| Q) - \frac{\chi^2(P' \| Q)}{2n} + O(n^{-3/2}).$$

Combining the two parts yields

$$I(K; \mathbf{Z}) = \beta \log n + (1 - \beta) \left(D_{\text{KL}}(P' \| Q) - \frac{\chi^2(P' \| Q)}{2n} \right) + O(n^{-3/2}),$$

as claimed. $\square$

Theorem 20. *In the canonical shuffle-only model, when $P \not\ll Q$, the mutual information between the target message and the shuffled output satisfies*

$$I(Y_1; \mathbf{Z}) = \sum_{y \in \mathbb{Y}_2} P(y) \log \frac{1}{P(y)} + (1 - \beta) \log \frac{1}{1 - \beta} + (1 - \beta) \frac{\sum_i \frac{p'_i - (p'_i)^2}{q_i}}{2n} + O(n^{-3/2}), \quad n \rightarrow \infty,$$

where $\beta = P(\mathbb{Y}_2)$, $P' = P|_{\mathbb{Y} \setminus \mathbb{Y}_2}$, and $\mathbb{Y}_2 = \{y \mid P(y) > 0 \text{ and } Q(y) = 0\}$.

Proof. Analogously to the proof of Theorem 19, we obtain

$$I(Y_1; \mathbf{Z}) = I(Y_1; \mathbf{Z} | T) + I(Y_1; T) = I(Y_1; \mathbf{Z} | T = 0) \Pr[T = 0] + I(Y_1; \mathbf{Z} | T = 1) \Pr[T = 1] + I(Y_1; T).$$

By definition,

$$I(Y_1; T) = H(Y_1) - H(Y_1 | T) = H(Y_1) - \beta H(Y_1 | T = 1) - (1 - \beta) H(Y_1 | T = 0).$$

When $T = 1$, the shuffle output directly reveals the value of Y_1 , thus $I(Y_1; \mathbf{Z} | T = 1) = H(Y_1 | T = 1)$. When $T = 0$, we again invoke Theorem 2 (the case $P' \ll Q$), giving

$$I(Y_1; \mathbf{Z} | T = 0) = \frac{\sum_i \frac{p'_i - (p'_i)^2}{q_i}}{2n} + O(n^{-3/2}).$$

By definition,

$$H(Y_1 | T = 0) = \sum_{y \in \mathbb{Y}_1} \frac{P(y)}{1 - \beta} \log \frac{1 - \beta}{P(y)}.$$

Algorithm 1 Post-processing function of privacy blanket, f^B

Meta parameter: $\text{LO}_j, j = 2, 3, \dots, n$

Input: Z^r

$Z \leftarrow Z^r$

$J \leftarrow \emptyset$

$S \leftarrow \emptyset$

for $i = 1, \dots, n$ **do**

if $Z^r[i] = \perp$ **then**

 Let j be a randomly and uniformly chosen element of $[2 : n] \setminus J$

$s \leftarrow_{\text{LO}_j} \mathbb{Y}$

$\triangleright$ Sample from LO_j

$J \leftarrow J \cup \{j\}$

$Z[i] = s$

end if

end for

return Z

Combining these parts yields

$$\begin{aligned}
 I(Y_1; Z) &= H(Y_1) - (1 - \beta)H(Y_1 \mid T = 0) + I(Y_1; Z \mid T = 0) \Pr[T = 0] \\
 &= \sum_{y \in \mathbb{Y}} P(y) \log \frac{1}{P(y)} - \sum_{y \in \mathbb{Y}_1} P(y) \log \frac{1 - \beta}{P(y)} + (1 - \beta) \frac{\sum_i \frac{p'_i - (p'_i)^2}{q_i}}{2n} + O(n^{-3/2}) \\
 &= \sum_{y \in \mathbb{Y}_2} P(y) \log \frac{1}{P(y)} + (1 - \beta) \log \frac{1}{1 - \beta} + (1 - \beta) \frac{\sum_i \frac{p'_i - (p'_i)^2}{q_i}}{2n} + O(n^{-3/2}). \square
 \end{aligned}$$

E The Post-Processing Function of Blanket Decomposition

The corresponding algorithm is shown in Algorithm 1.

F Details in Proof of Lemma 1

Fact 1. $\Pr(X_1 = x \mid C^{(s)}) \propto P(x) T_x^{(s)}$.

Proof. Recall that $X_1 \sim P$, $Y_1 \mid X_1 = x \sim \mathcal{R}_x$, and $M^{(s)} \sim \text{Mult}(s; Q)$ is independent of (X_1, Y_1) . We observe the histogram of size $s+1$:

$$C^{(s)} = e_{Y_1} + M^{(s)}, \quad \text{i.e., } C_y^{(s)} = \mathbb{1}\{Y_1 = y\} + M_y^{(s)}, \quad \sum_y C_y^{(s)} = s + 1.$$

Fix any histogram $c = (c_y)_{y \in \mathbb{Y}}$ with total count $s + 1$. By Bayes' rule,

$$\Pr(X_1 = x \mid C^{(s)} = c) \propto P(x) \Pr(C^{(s)} = c \mid X_1 = x).$$

We compute the likelihood by conditioning on Y_1 :

$$\Pr(C^{(s)} = c \mid X_1 = x) = \sum_{y \in \mathbb{Y}} \Pr(Y_1 = y \mid X_1 = x) \Pr(M^{(s)} = c - e_y) = \sum_y \mathcal{R}_x(y) \Pr(M^{(s)} = c - e_y),$$

where the summand is zero when $c_y = 0$. The multinomial probability mass function is

$$\Pr(M^{(s)} = m) = \frac{s!}{\prod_z m_z!} \prod_z Q(z)^{m_z}, \quad \sum_z m_z = s.$$

For each y with $c_y \geq 1$,

$$\Pr(M^{(s)} = c - e_y) = \frac{s!}{\prod_z (c_z - \mathbb{1}\{z=y\})!} \prod_z Q(z)^{c_z - \mathbb{1}\{z=y\}}.$$

Using $(c_y - 1)! = c_y! / c_y$ and $Q(z)^{c_z - \mathbb{1}\{z=y\}} = Q(z)^{c_z} Q(y)^{-1}$, we can separate the factor independent of y :

$$\Pr(M^{(s)} = c - e_y) = \left[\frac{s!}{\prod_z c_z!} \prod_z Q(z)^{c_z} \right] \frac{c_y}{Q(y)}.$$

Thus,

$$\Pr(C^{(s)} = c \mid X_1 = x) = \left[\frac{s!}{\prod_z c_z!} \prod_z Q(z)^{c_z} \right] \sum_y \mathcal{R}_x(y) \frac{c_y}{Q(y)}.$$

The bracketed term does not depend on x . Hence,

$$\Pr(X_1 = x \mid C^{(s)} = c) \propto P(x) \sum_y c_y \frac{\mathcal{R}_x(y)}{Q(y)} = P(x) T_x^{(s)},$$

where we define

$$T_x^{(s)} := \sum_y c_y w_x(y), \quad w_x(y) := \frac{\mathcal{R}_x(y)}{Q(y)}.$$

Finally, replacing c by the random histogram $C^{(s)}$ yields

$$\Pr(X_1 = x \mid C^{(s)}) \propto P(x) T_x^{(s)}, \quad T_x^{(s)} = \sum_y C_y^{(s)} w_x(y). \quad \square$$

Lemma 5. *Let P be a probability mass function on a finite alphabet $\mathbb{X}$ and let*

$$\pi(x) = \frac{P(x)(1 + \varepsilon_x)}{1 + \bar{\varepsilon}}, \quad \bar{\varepsilon} := \sum_{u \in \mathbb{X}} P(u) \varepsilon_u,$$

with $\sum_x P(x) \varepsilon_x$ finite and $1 + \bar{\varepsilon} > 0$, $1 + \varepsilon_x > 0$ for all x . Then, for $\|\varepsilon\|_\infty$ sufficiently small, we have the second-order expansion

$$D(\pi \| P) = \frac{1}{2} \sum_{x \in \mathbb{X}} P(x) (\varepsilon_x - \bar{\varepsilon})^2 + R(\varepsilon), \quad (35)$$

with a remainder satisfying

$$|R(\varepsilon)| \leq K \|\varepsilon\|_3^3, \quad \|\varepsilon\|_3 := \left(\sum_{x \in \mathbb{X}} P(x) |\varepsilon_x|^3 \right)^{1/3}, \quad (36)$$

where K is a constant.

Proof. By definition,

$$D(\pi \| P) = \sum_x \pi(x) \log \frac{\pi(x)}{P(x)} = \sum_x \frac{P(x)(1 + \varepsilon_x)}{1 + \bar{\varepsilon}} \left(\log(1 + \varepsilon_x) - \log(1 + \bar{\varepsilon}) \right).$$

Write $a_x := \varepsilon_x$ and $b := \bar{\varepsilon}$ for brevity. Using the Taylor expansion

$$\log(1 + u) = u - \frac{u^2}{2} + r_3(u), \quad |r_3(u)| \leq C |u|^3 \quad \text{for } |u| \leq u_0,$$

we obtain, for all x with $|a_x|, |b| \leq u_0$,

$$\log(1 + a_x) - \log(1 + b) = (a_x - b) - \frac{1}{2}(a_x^2 - b^2) + (r_3(a_x) - r_3(b)).$$

Also expand the prefactor:

$$\frac{1 + a_x}{1 + b} = 1 + (a_x - b) + \rho_x, \quad \rho_x := -\frac{(a_x - b)b}{1 + b},$$

so that $|\rho_x| \leq C'(|a_x| + |b|)|b|$ for $|b| \leq u_0$. Hence

$$\begin{aligned} D(\pi \| P) &= \sum_x P(x) \left(1 + (a_x - b) + \rho_x \right) \left[(a_x - b) - \frac{1}{2}(a_x^2 - b^2) + r_3(a_x) - r_3(b) \right] \\ &= \sum_x P(x) \left[(a_x - b) - \frac{1}{2}(a_x^2 - b^2) \right] + \sum_x P(x)(a_x - b) \left[(a_x - b) - \frac{1}{2}(a_x^2 - b^2) \right] \\ &\quad + \sum_x P(x)\rho_x \left[(a_x - b) - \frac{1}{2}(a_x^2 - b^2) \right] + \sum_x P(x) \left(1 + (a_x - b) + \rho_x \right) (r_3(a_x) - r_3(b)). \end{aligned}$$

The *linear* term vanishes:

$$\sum_x P(x)(a_x - b) = \sum_x P(x)a_x - b = 0.$$

For the *quadratic* contribution, note that

$$-\frac{1}{2} \sum_x P(x)(a_x^2 - b^2) = -\frac{1}{2} \sum_x P(x)a_x^2 + \frac{1}{2}b^2, \quad \sum_x P(x)(a_x - b)^2 = \sum_x P(x)a_x^2 - b^2.$$

Therefore,

$$-\frac{1}{2} \sum_x P(x)(a_x^2 - b^2) + \sum_x P(x)(a_x - b)^2 = \frac{1}{2} \sum_x P(x)(a_x - b)^2.$$

All the remaining pieces are cubic (or higher) in $\{a_x\}$ and b :

- $\sum_x P(x)(a_x - b) \cdot (-\frac{1}{2}(a_x^2 - b^2))$ is $O(\sum P|a|^3 + |b|^3)$;
- $\sum_x P(x)\rho_x \left[(a_x - b) - \frac{1}{2}(a_x^2 - b^2) \right]$ is bounded by a constant times $\sum_x P(x)(|a_x| + |b|)|b|(|a_x - b| + |a_x|^2 + |b|^2)$, hence $O(\sum P|a|^3 + |b|^3)$;
- $\sum_x P(x) \left(1 + (a_x - b) + \rho_x \right) (r_3(a_x) - r_3(b))$ is $O(\sum P|a_x|^3 + |b|^3)$ by the uniform cubic bound on r_3 .

Since $b = \sum_x P(x) a_x$, the triangle inequality gives $|b| \leq \sum_x P(x) |a_x|$. Applying Jensen's inequality to the convex function $\varphi(t) = |t|^3$ yields

$$|b|^3 \leq \left(\sum_x P(x) |a_x| \right)^3 \leq \sum_x P(x) |a_x|^3.$$

and thus each cubic contribution is bounded by a constant multiple of $\sum_x P(x) |a_x|^3 = \|\varepsilon\|_3^3$. Collecting terms yields (35)–(36). $\square$

Fact 2. Let $M^{(s)} = (M_y^{(s)})_{y \in \mathbb{Y}} \sim \text{Mult}(s; Q)$ with $Q(y) > 0$ and

$$c := M^{(s)} - sQ, \quad c_y := M_y^{(s)} - sQ(y).$$

Then

$$\mathbb{E}[c] = 0, \quad \mathbb{E}[cc^\top] = \text{Cov}(M^{(s)}) = s(\text{Diag}(Q) - QQ^\top).$$

Proof. Write the multinomial vector as a sum of i.i.d. one-hot indicators. Let $Y_1, \dots, Y_s$ be i.i.d. on $\mathbb{Y}$ with $\Pr(Y_t = y) = Q(y)$ and define

$$\mathbf{1}_t(y) := \mathbb{1}\{Y_t = y\}, \quad M_y^{(s)} = \sum_{t=1}^s \mathbf{1}_t(y).$$

Then

$$\mathbb{E}[M_y^{(s)}] = \sum_{t=1}^s \mathbb{E}[\mathbf{1}_t(y)] = \sum_{t=1}^s Q(y) = sQ(y),$$

so $\mathbb{E}[c_y] = \mathbb{E}[M_y^{(s)}] - sQ(y) = 0$, i.e. $\mathbb{E}[c] = 0$.

For the second moments, for $a, b \in \mathbb{Y}$,

$$\text{Cov}(M_a^{(s)}, M_b^{(s)}) = \sum_{t=1}^s \text{Cov}(\mathbf{1}_t(a), \mathbf{1}_t(b))$$

because different trials $t \neq t'$ are independent. Now

$$\text{Cov}(\mathbf{1}_t(a), \mathbf{1}_t(b)) = \begin{cases} \text{Var}(\mathbf{1}_t(a)) = Q(a)(1 - Q(a)), & a = b, \\ \mathbb{E}[\mathbf{1}_t(a)\mathbf{1}_t(b)] - \mathbb{E}[\mathbf{1}_t(a)]\mathbb{E}[\mathbf{1}_t(b)] = 0 - Q(a)Q(b) = -Q(a)Q(b), & a \neq b, \end{cases}$$

since $\mathbf{1}_t(a)\mathbf{1}_t(b) = 0$ a.s. for $a \neq b$ (mutually exclusive one-hot). Therefore

$$\text{Cov}(M_a^{(s)}, M_b^{(s)}) = s(Q(a)\mathbf{1}\{a = b\} - Q(a)Q(b)).$$

Stacking these entries gives the covariance matrix

$$\text{Cov}(M^{(s)}) = s(\text{Diag}(Q) - QQ^\top).$$

Finally, because $c = M^{(s)} - \mathbb{E}[M^{(s)}]$ is the centered vector, we have

$$\mathbb{E}[cc^\top] = \text{Cov}(M^{(s)}),$$

hence

$$\mathbb{E}[cc^\top] = s(\text{Diag}(Q) - QQ^\top). \quad \square$$

Fact 3 (Variance of a Linear Statistic via Covariance Matrix). *Let $\mathbb{Y} = \{y_1, \dots, y_m\}$ and define, for each x ,*

$$w_x := (w_x(y_1), \dots, w_x(y_m))^\top \in \mathbb{R}^m, \quad \bar{w} := \mathbb{E}_{x \sim P}[w_x].$$

Let the covariance matrix

$$\Sigma_w := \text{Cov}_{x \sim P}(w_x) = \mathbb{E}_x[(w_x - \bar{w})(w_x - \bar{w})^\top].$$

For any (possibly random) vector $c = (c_{y_1}, \dots, c_{y_m})^\top \in \mathbb{R}^m$ that is independent of x , define the linear statistic

$$A_x := \sum_y c_y w_x(y) = c^\top w_x.$$

(1) (**Conditional variance identity**) *For any fixed c ,*

$$\text{Var}_{x \sim P}(A_x) = c^\top \Sigma_w c.$$

(2) (**Taking expectation over c**) *If c is random and independent of x , then*

$$\mathbb{E}[\text{Var}_{x \sim P}(A_x)] = \mathbb{E}[c^\top \Sigma_w c] = \text{Tr}(\Sigma_w \mathbb{E}[c c^\top]).$$

Proof. Since $\mathbb{E}_x[A_x] = c^\top \bar{w}$, we have

$$A_x - \mathbb{E}_x[A_x] = c^\top (w_x - \bar{w}).$$

Thus,

$$\text{Var}_x(A_x) = \mathbb{E}_x[(c^\top (w_x - \bar{w}))^2] = c^\top \mathbb{E}_x[(w_x - \bar{w})(w_x - \bar{w})^\top] c = c^\top \Sigma_w c.$$

If c is random but independent of x , taking expectation yields

$$\mathbb{E}[\text{Var}_x(A_x)] = \mathbb{E}[c^\top \Sigma_w c] = \text{Tr}(\Sigma_w \mathbb{E}[c c^\top]),$$

using $\text{Tr}(AB) = \text{Tr}(BA)$ and linearity of trace. □

Lemma 6.

$$\mathbb{E}[\text{Var}_{x \sim P}(A_x^{(s)})] = s(\overline{\chi^2} - \chi^2(\bar{P} \| Q)) + O(1).$$

Proof. Let $\mathbf{c} \in \mathbb{R}^{|\mathbb{Y}|}$ be defined by $c_y := M_y^{(s)} - sQ(y)$. Then $\mathbb{E}[\mathbf{c}] = \mathbf{0}$ and $\mathbb{E}[\mathbf{c}\mathbf{c}^\top] = s(\text{Diag}(Q) - \mathbf{Q}\mathbf{Q}^\top)$, where $\text{Diag}(Q)$ is a $|\mathbb{Y}| \times |\mathbb{Y}|$ matrix with diagonal $\text{Diag}(Q)_{ii} = Q(i)$, and $\mathbf{Q}$ is a column vector with $\mathbf{Q}_i = Q(i)$ (Fact 2 in Appendix F).

Define the covariance matrix (with respect to $x \sim P$)

$$\Sigma_w := \text{Cov}_{x \sim P}(w_x(\cdot)), \quad (\Sigma_w)_{ab} = \text{Cov}_x(w_x(a), w_x(b)).$$

For fixed $(M^{(s)}, Y_1)$,

$$\text{Var}_{x \sim P}(A_x^{(s)}) = \text{Var}_{x \sim P}\left(\sum_y c_y w_x(y)\right) + \text{Var}_{x \sim P}(w_x(Y_1)) + 2 \text{Cov}_{x \sim P}\left(\sum_y c_y w_x(y), w_x(Y_1)\right).$$

Taking expectation in $(M^{(s)}, Y_1)$, the cross term vanishes since $\mathbb{E}[c_y] = 0$ and $\mathbf{c}$ is independent of (x, Y_1) . Therefore

$$\mathbb{E}[\text{Var}_{x \sim P}(A_x^{(s)})] = \mathbb{E}\left[\text{Var}_{x \sim P}\left(\sum_y c_y w_x(y)\right)\right] + \mathbb{E}[\text{Var}_{x \sim P}(w_x(Y_1))].$$

The first term equals (Fact 3 in Appendix F)

$$\mathbb{E}[\mathbf{c}^\top \Sigma_w \mathbf{c}] = \text{Tr}(\Sigma_w \mathbb{E}[\mathbf{c}\mathbf{c}^\top]) = s \text{Tr}(\Sigma_w (\text{Diag}(Q) - \mathbf{Q}\mathbf{Q}^\top)) = s \text{Tr}(\Sigma_w \cdot \text{Diag}(Q)) = s \sum_y Q(y) \text{Var}_{x \sim P}(w_x(y)),$$

because $\text{Tr}(\mathbf{Q}\mathbf{Q}^\top \Sigma_w) = \text{Var}_{x \sim P}(\sum_y Q(y) w_x(y)) = 0$, as $\sum_y Q(y) w_x(y) = \sum_y \mathcal{R}_x(y) = 1$ for all x . Now observe that

$$\begin{aligned} \sum_y Q(y) \text{Var}_x(w_x(y)) &= \sum_y Q(y) \mathbb{E}_x[w_x^2(y)] - \sum_y Q(y) (\mathbb{E}_x[w_x(y)])^2 \\ &= \mathbb{E}_x \left[\sum_y \frac{\mathcal{R}_x(y)^2}{Q(y)} \right] - \sum_y \frac{\bar{P}(y)^2}{Q(y)} \\ &= \bar{\chi}^2 - \chi^2(\bar{P} \| Q). \end{aligned}$$

The second term $\mathbb{E}[\text{Var}_x(w_x(Y_1))]$ is $O(1)$: the marginal of Y_1 is $\bar{P}$, and on a finite alphabet with $Q(y) > 0$, the family $\{w_x(y)\}$ is uniformly bounded. Thus

$$\mathbb{E}[\text{Var}_{x \sim P}(A_x^{(s)})] = s(\bar{\chi}^2 - \chi^2(\bar{P} \| Q)) + O(1). \quad \square$$